

Visualization and interpretation of high-throughput genomics data

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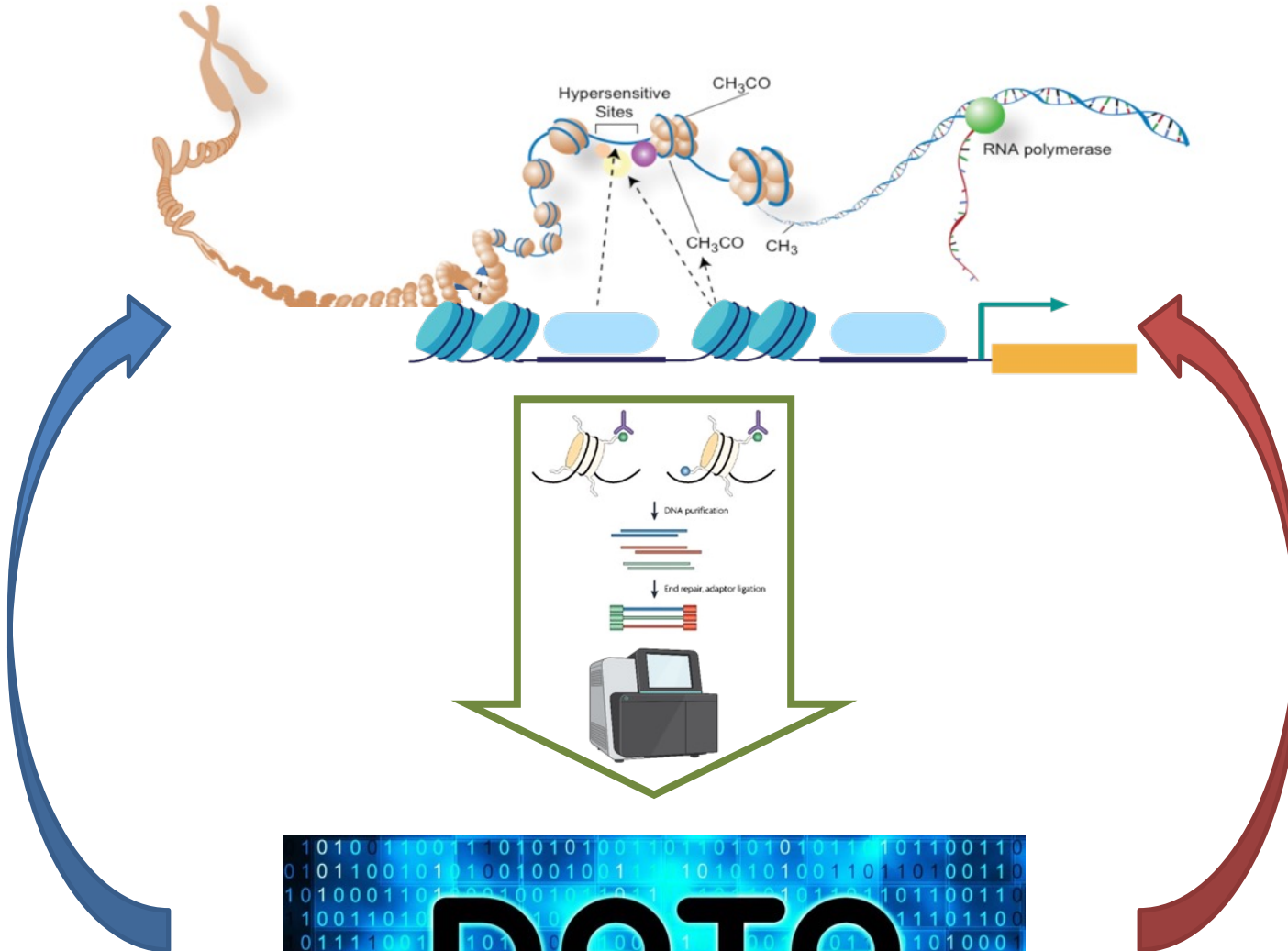


My lab develops computational methods and uses computational approaches to study epigenetics and transcriptional regulation



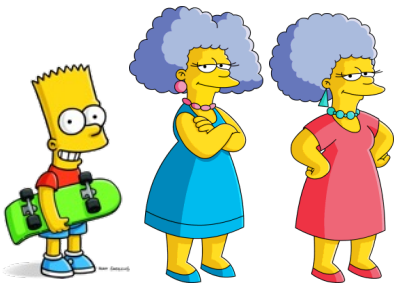
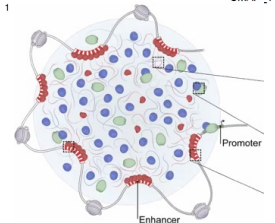
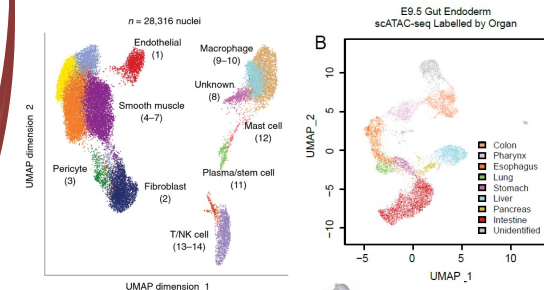
Method development

- Stat/math/physics /machine learning models
- Algorithms
- Software tools



Data-inspired discovery

- Multi-omics data integration
- New models
- New discoveries



LET'S SOLVE THIS PROBLEM BY
USING THE BIG DATA NONE
OF US HAVE THE SLIGHTEST
IDEA WHAT TO DO WITH

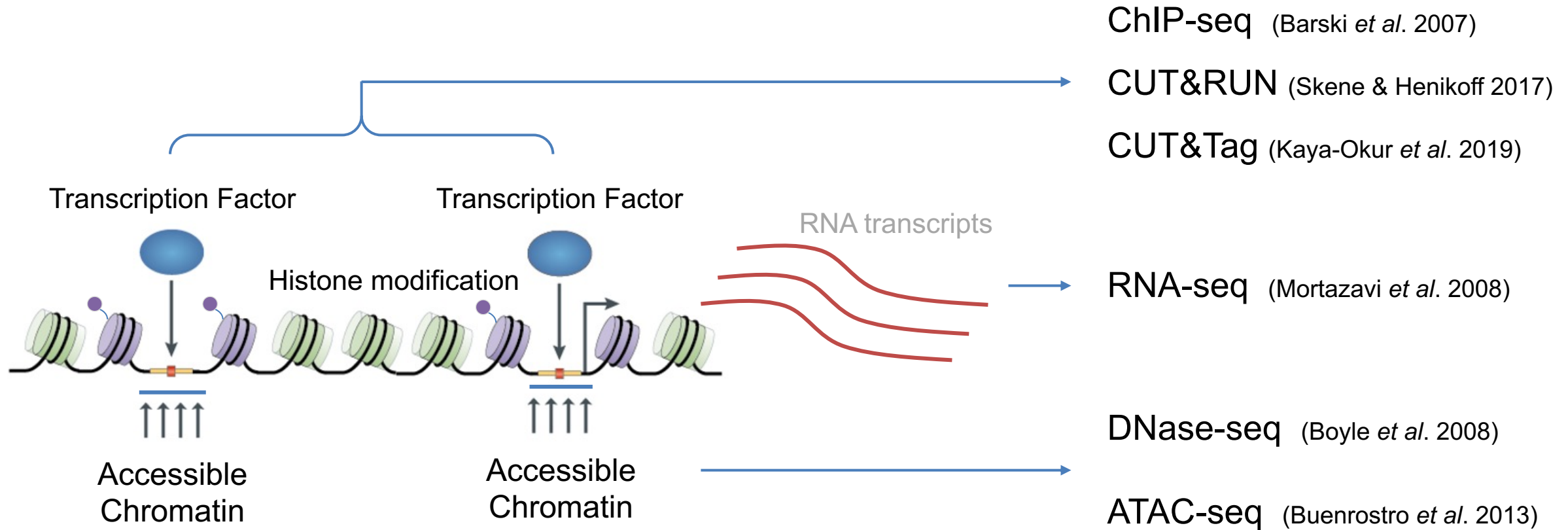


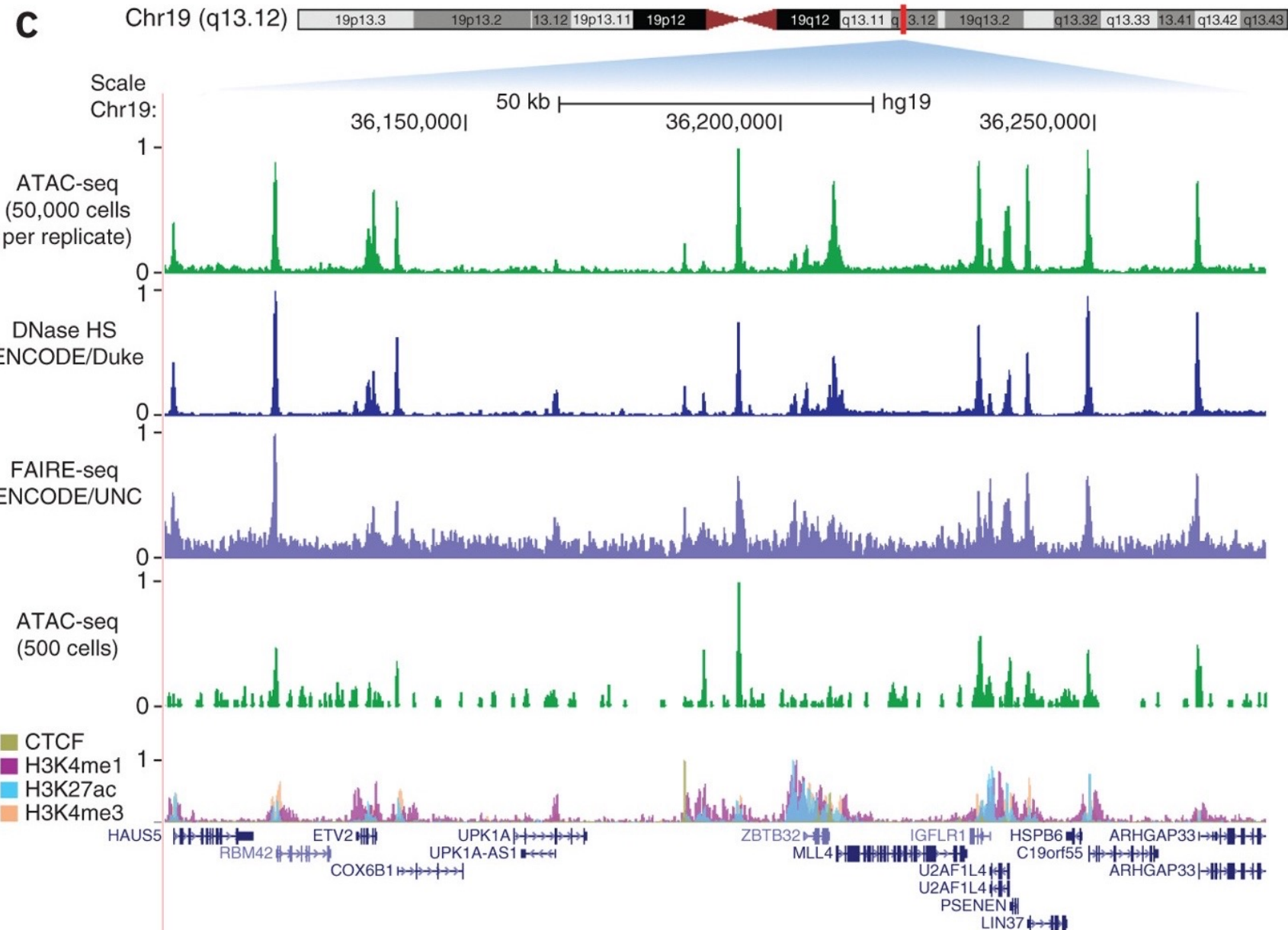
© marketoonist.com

Learning Objectives

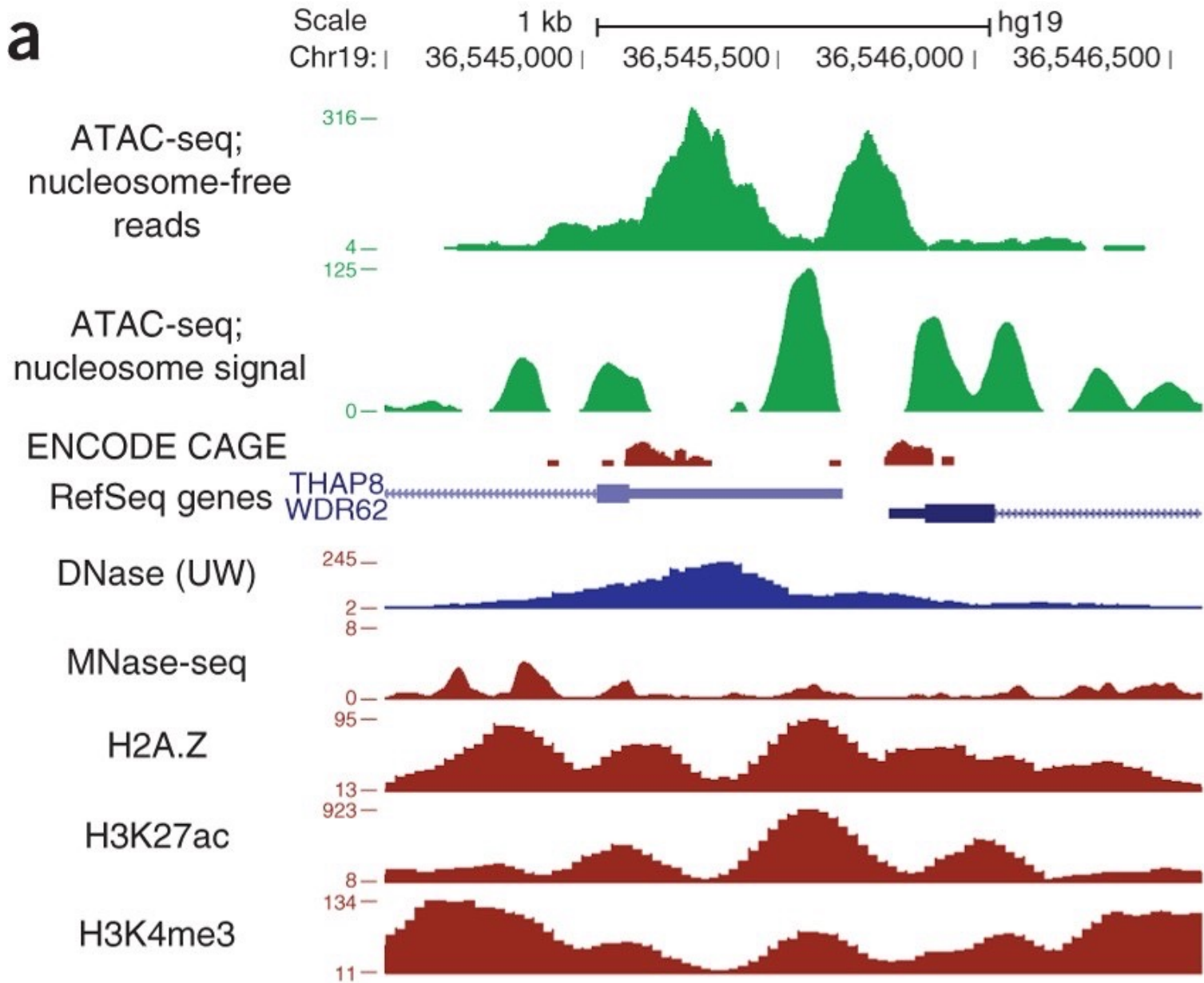
- Understand common types of plots for presenting high-throughput genomics data
- Understand essential elements of effective genomics data visualization
- Learn key principles and tips for data presentation

High-throughput genomics technologies enable genome-wide measurements

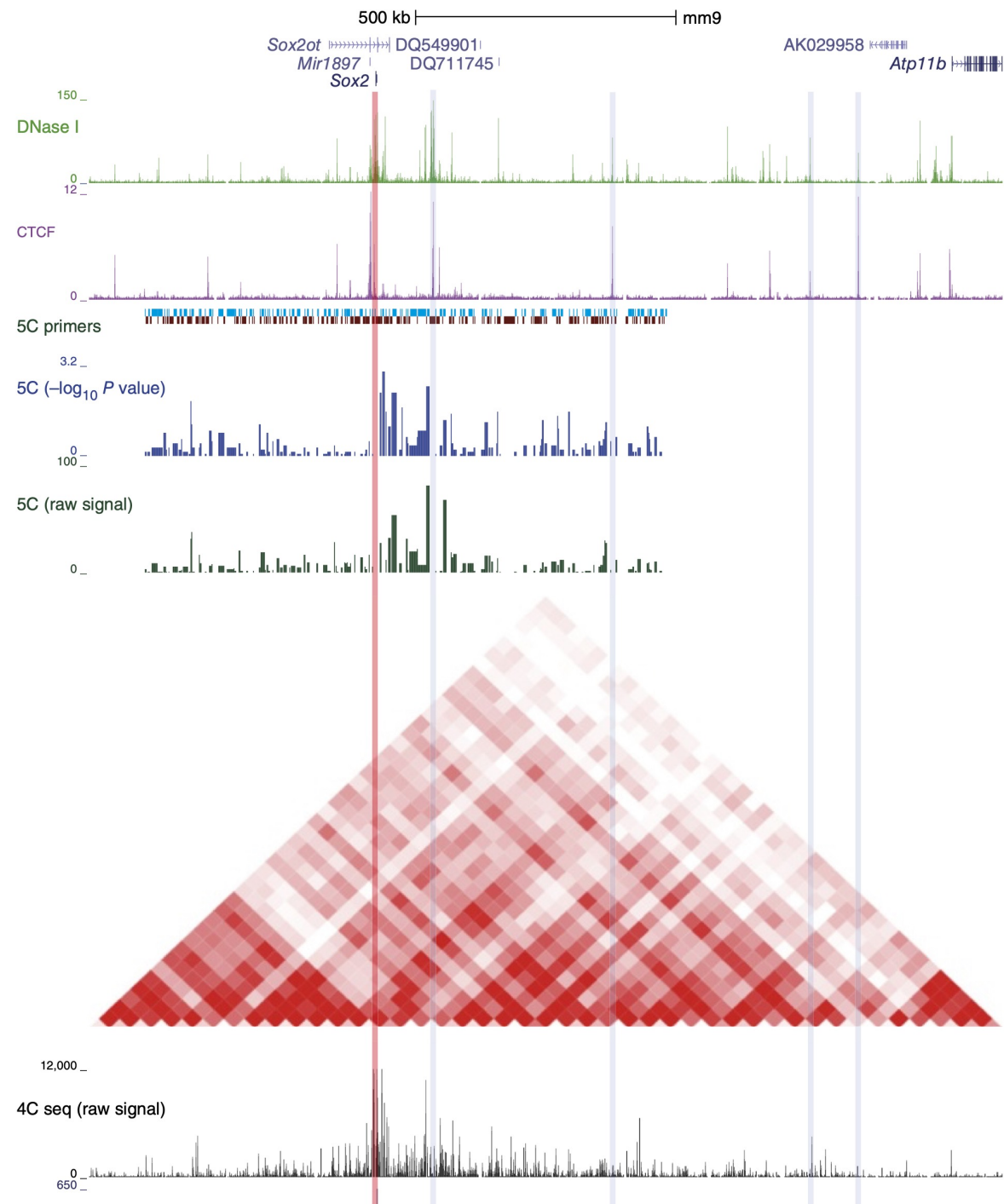




Buenrostro *et al.* Nat Methods 2013

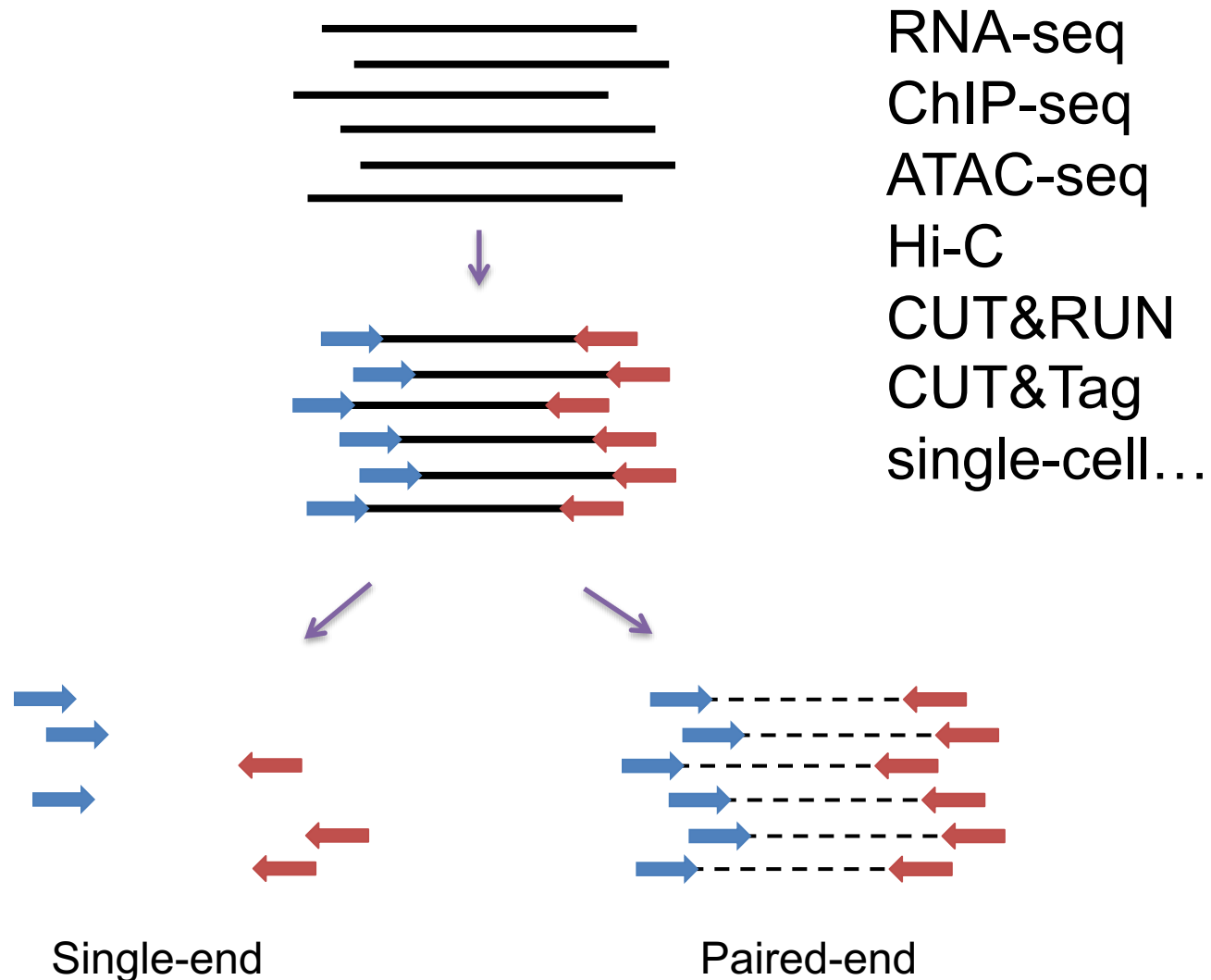
a

Buenrostro *et al.* *Nat Methods* 2013

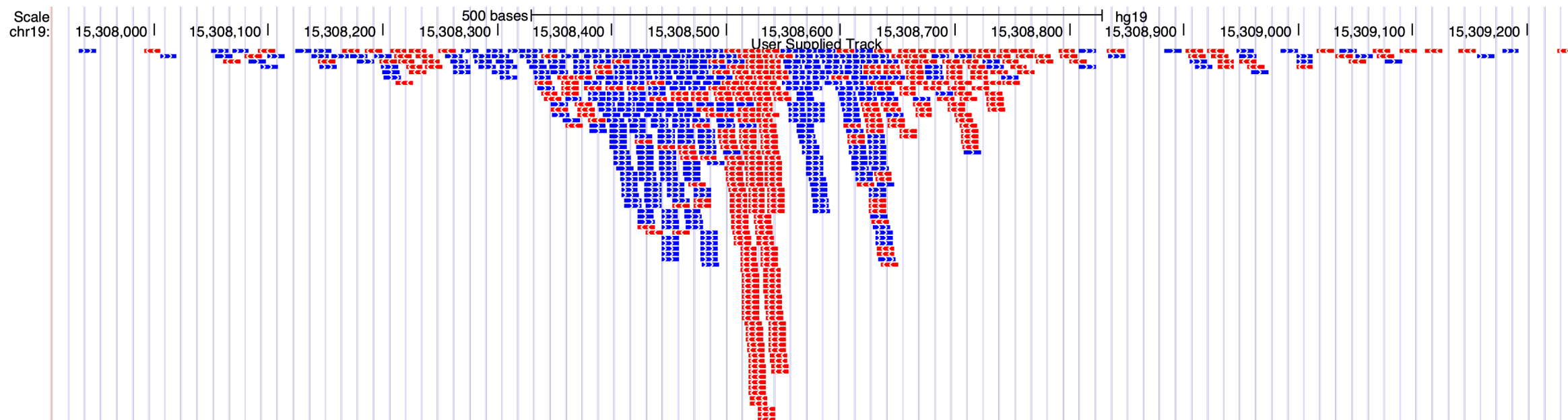


Davies *et al.* Nat Methods 2017

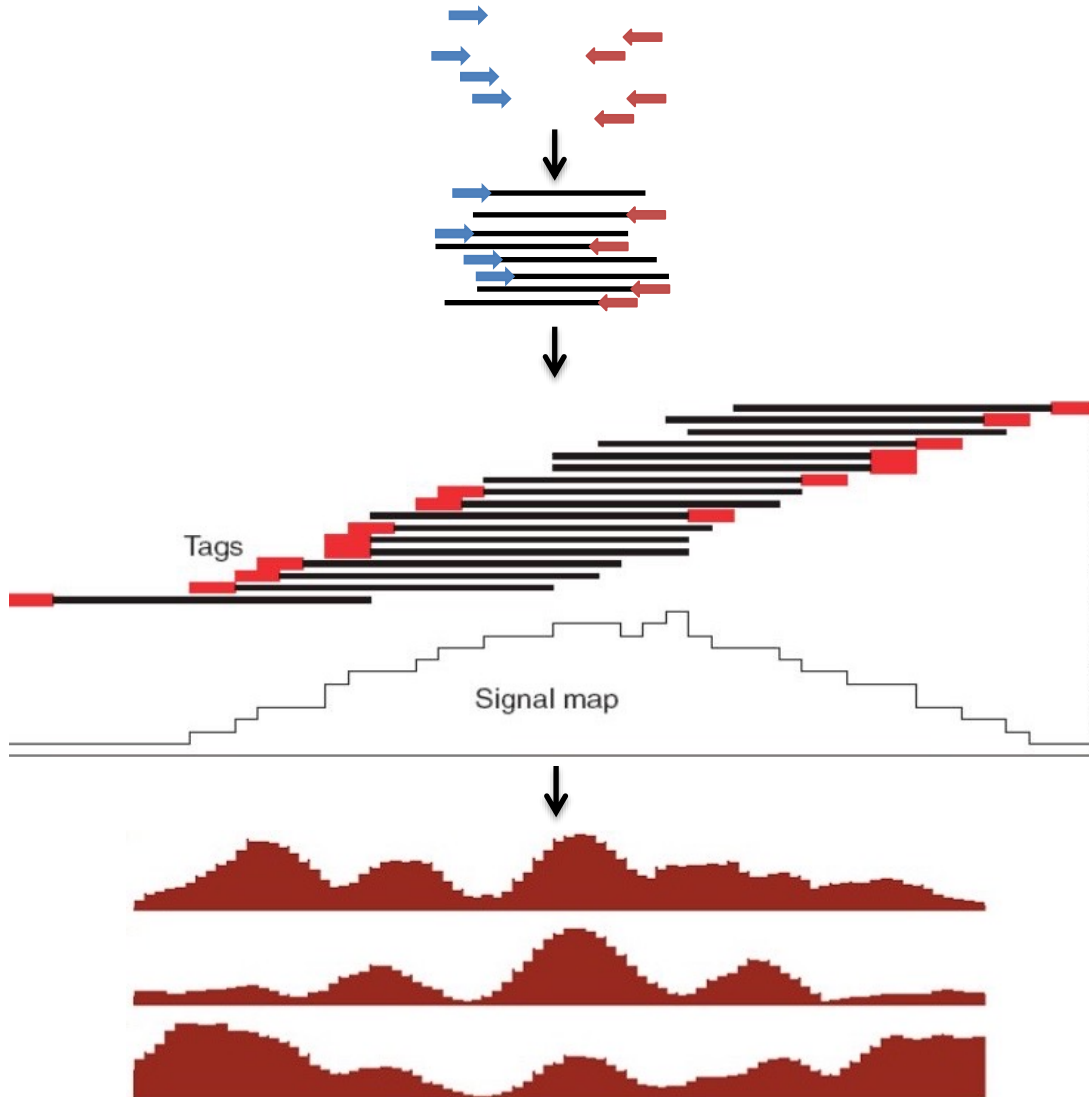
High-throughput short-read sequencing (Illumina)



Original sequence reads are not easy to visualize



Signal tracks are sequence fragments piled up



- bedGraph:

chr4	10344200	10344250	5
chr4	10344250	10344300	10
chr4	10344300	10344350	25
chr4	10344350	10344400	15
chr4	10344400	10344450	8

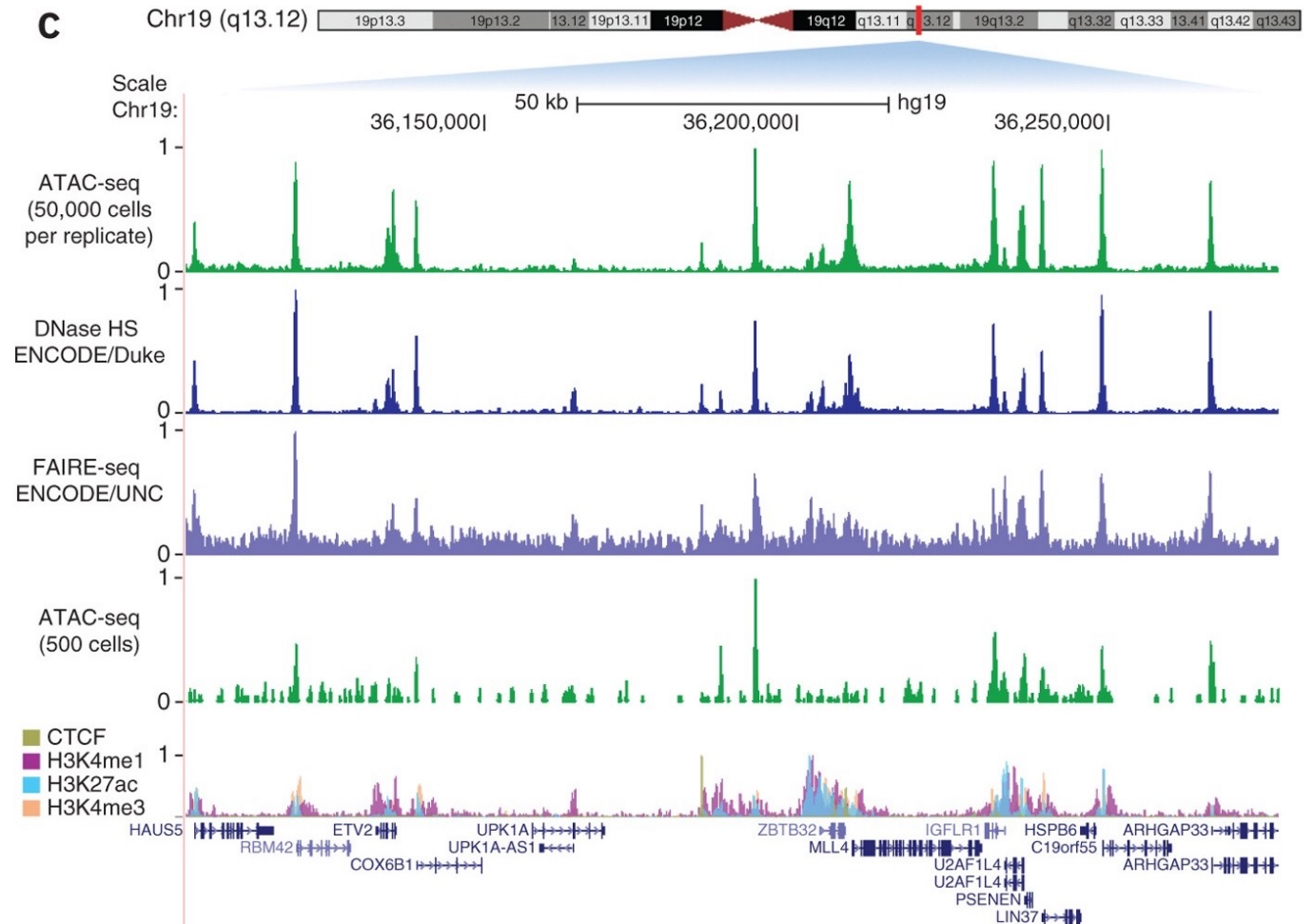
- wiggle:

```
track type=wiggle_0
variableStep chrom=chr4 span=50
10344200 5
10344250 10
10344300 25
10344350 15
10344400 8
```

- bigWig: indexed binary format

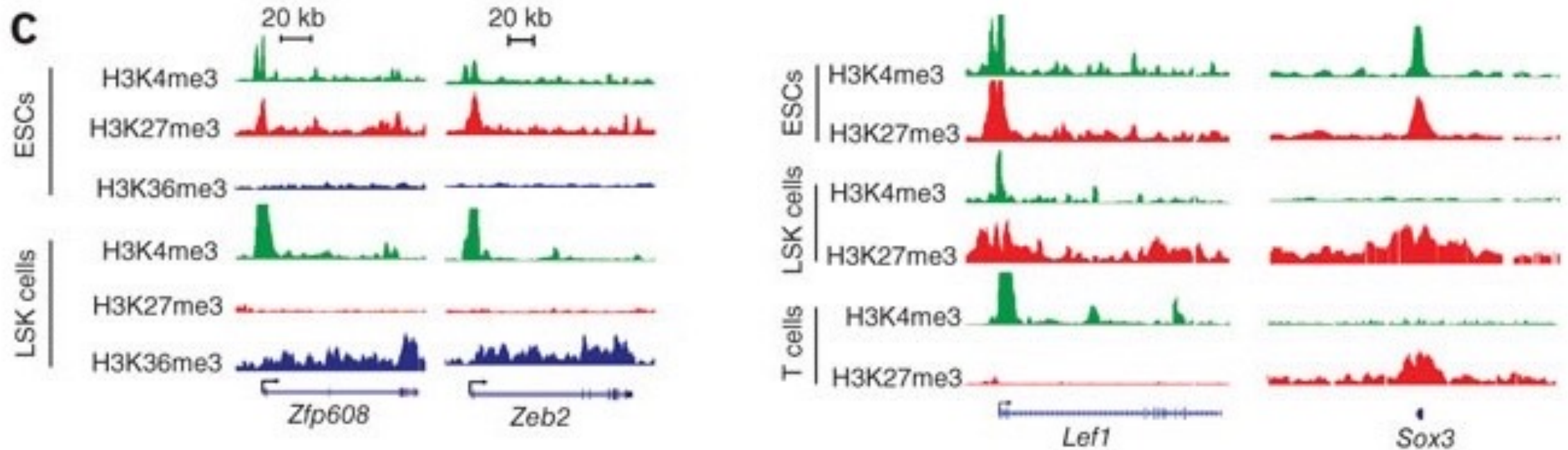
Essential elements in genome browser tracks

- Chromosomal locations
- Track label
- Track scale
 - x: resolution?
 - y: normalization?

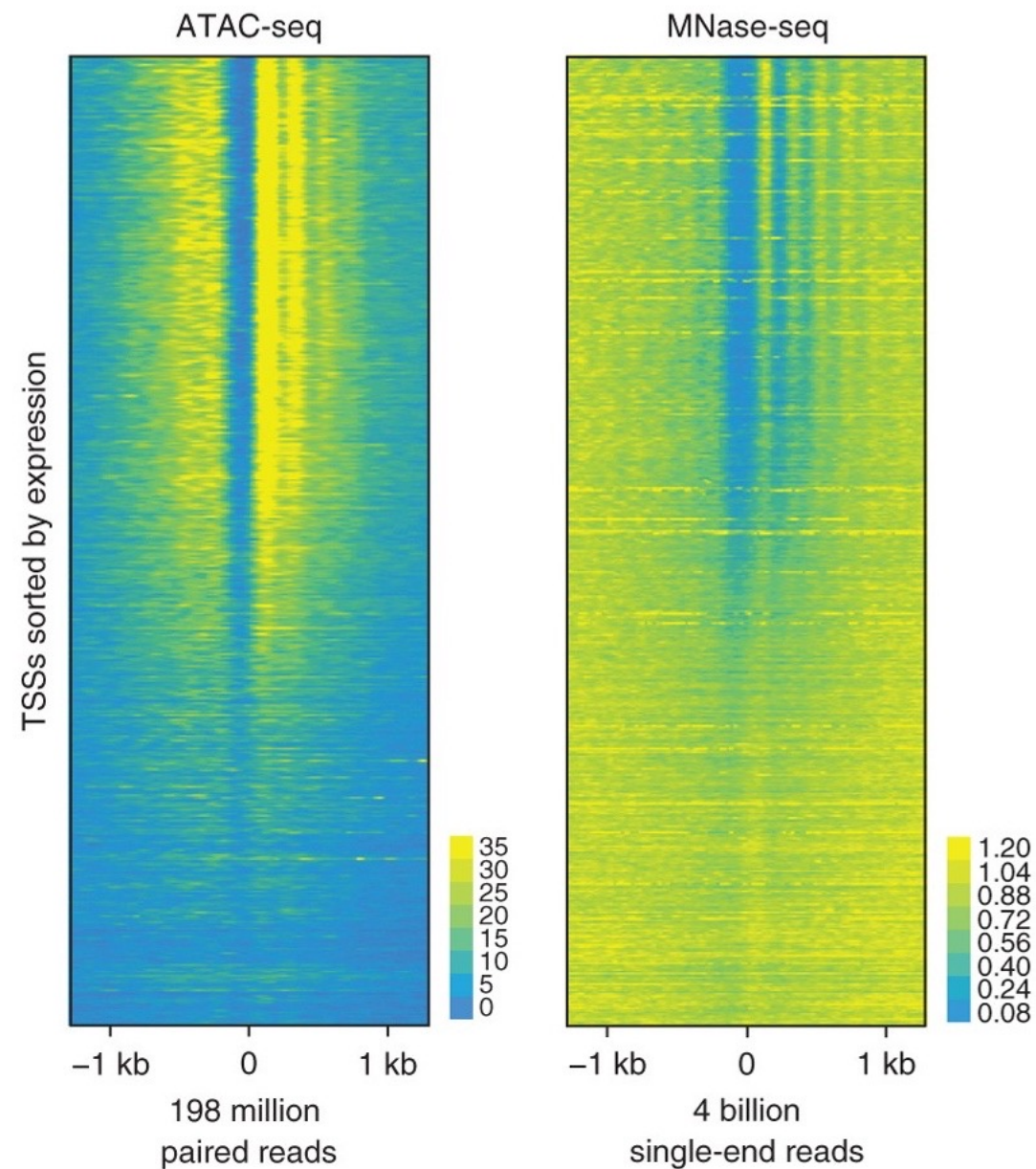
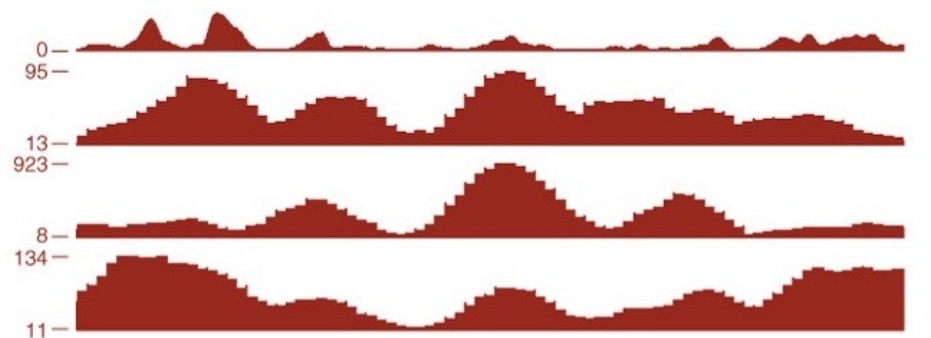


Buenrostro *et al.* Nat Methods 2013

How to integrate patterns observed on signal tracks?



Stacked “ripple” heatmap

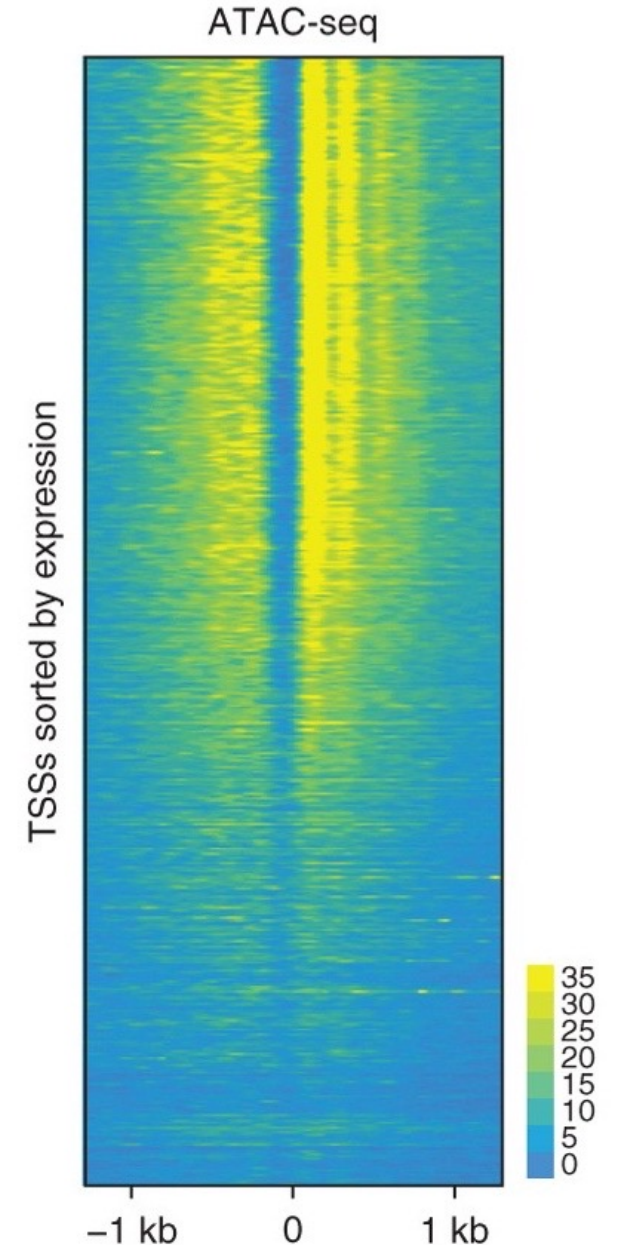


Buenrostro *et al.* Nat Methods 2013

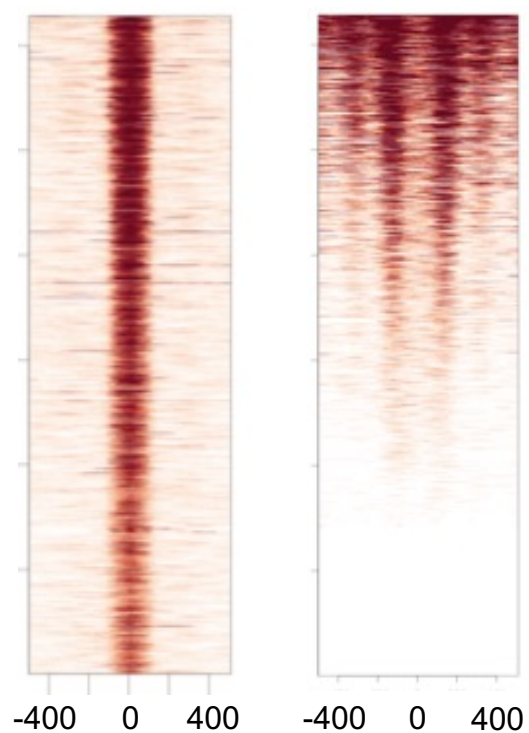


Essential elements in a ripple heatmap

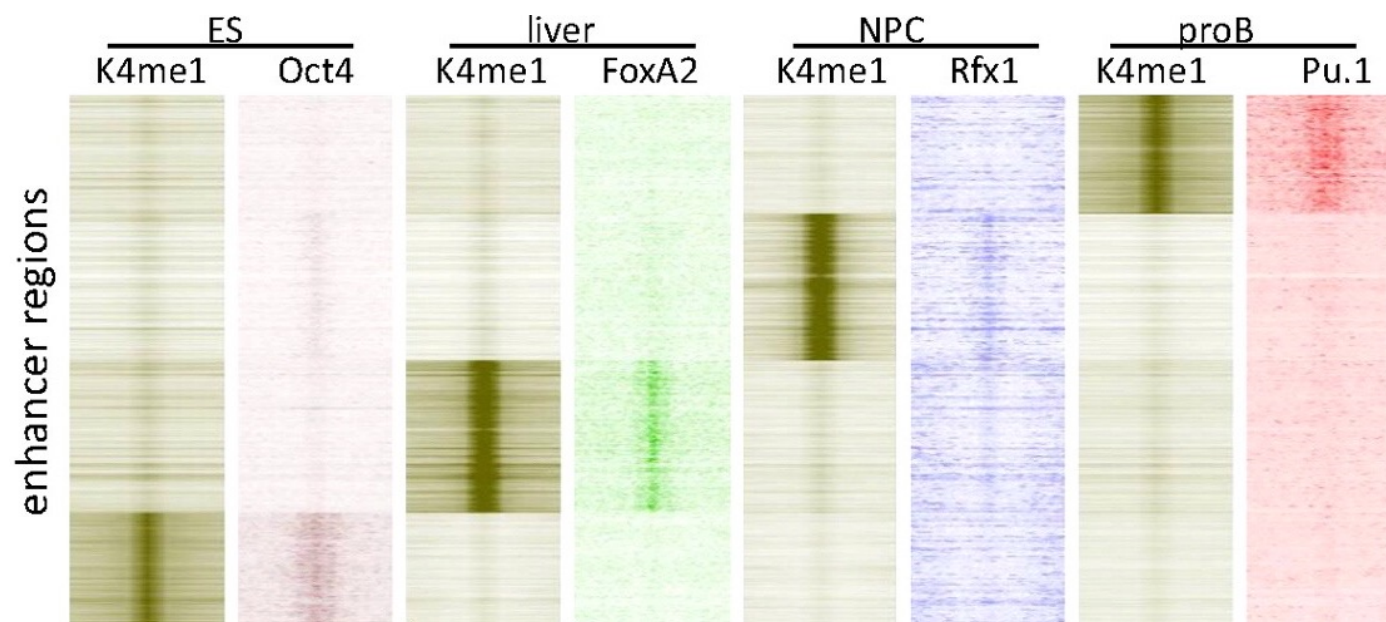
- Heatmap presents 3-dimensional data
- x: What loci/anchor is each row? Range?
- y: What are the rows? How many? How are they ranked?
- h: Data title/label (what signal?) color scale?



Multiple datasets visualization by ripple heatmap

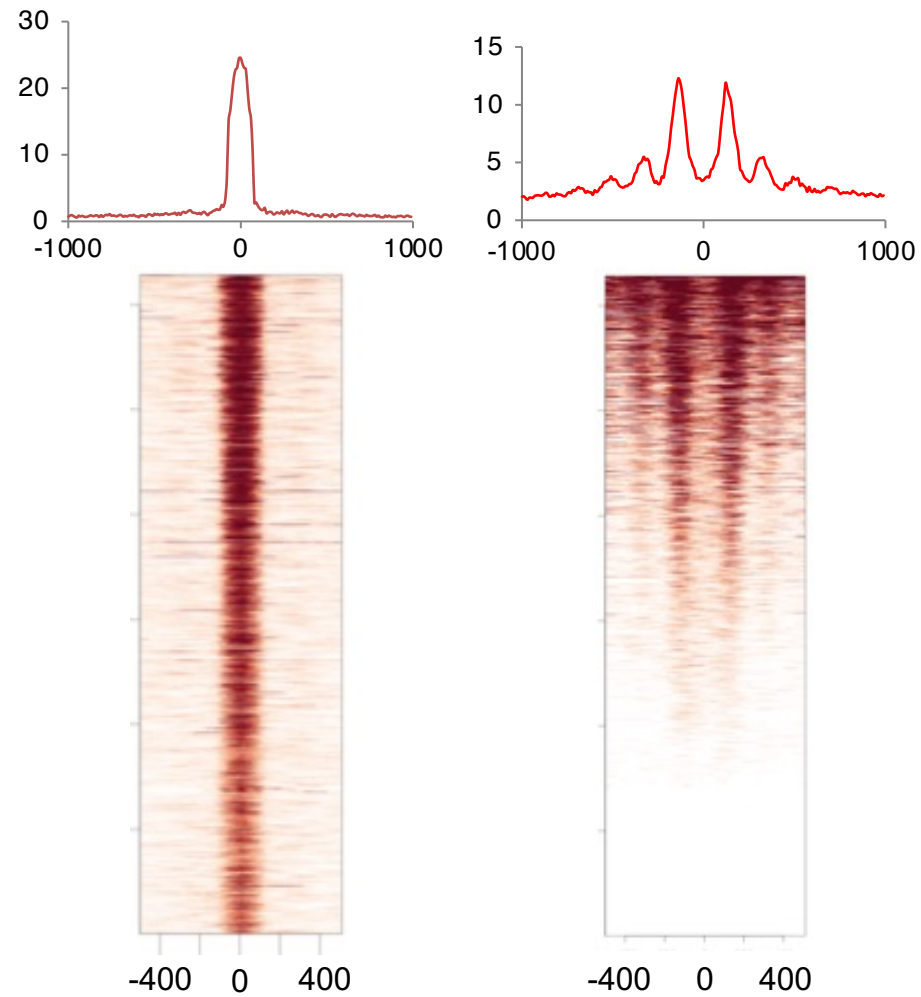


Luyten *et al. Genes Dev* 2014



Creyghton *et al. PNAS* 2010

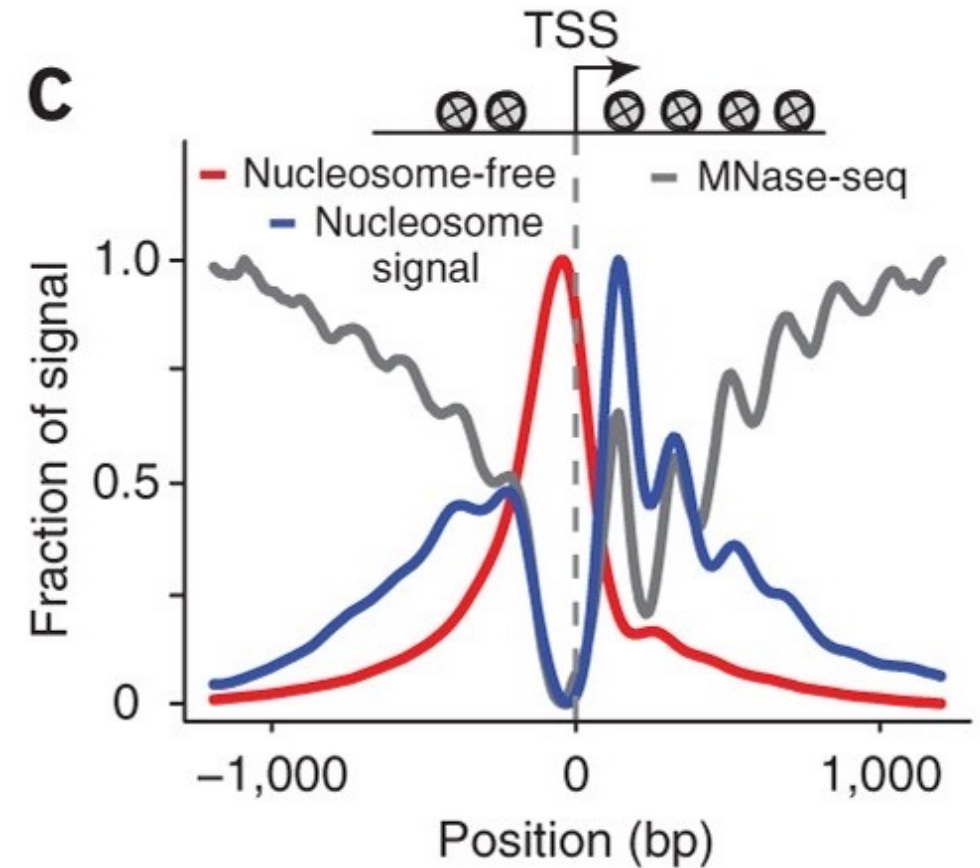
Composite plots



Luyten *et al.* *Genes Dev* 2014

Essential elements in a composite plot

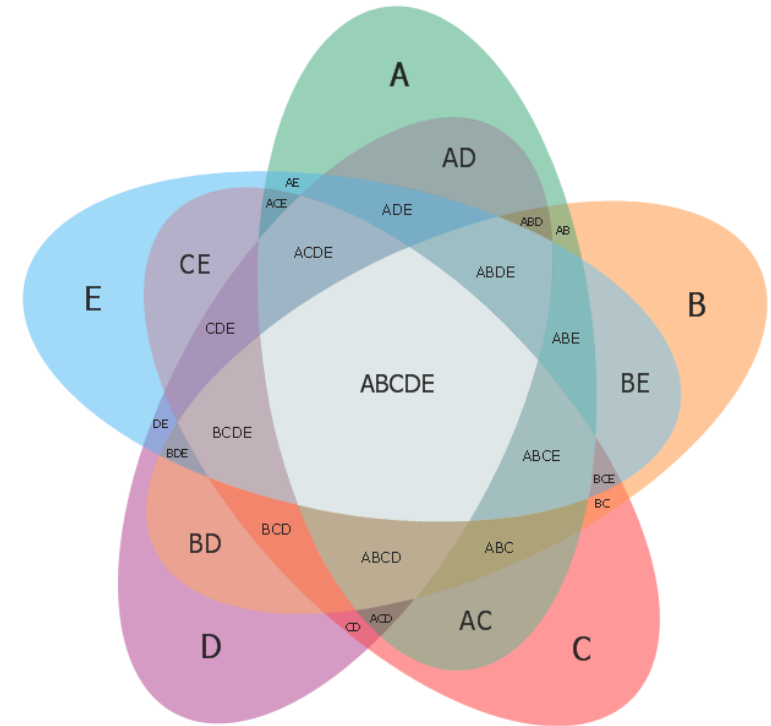
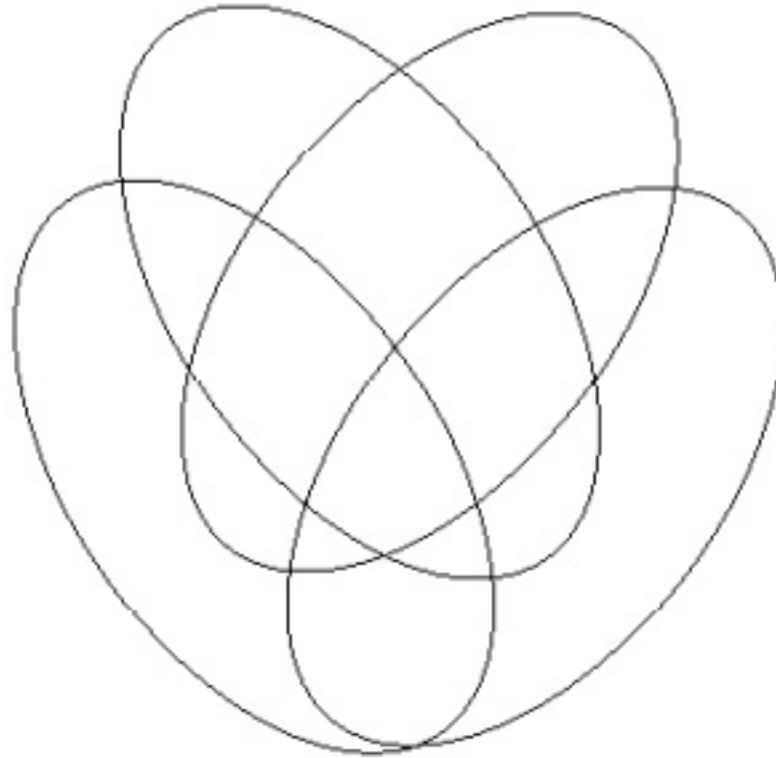
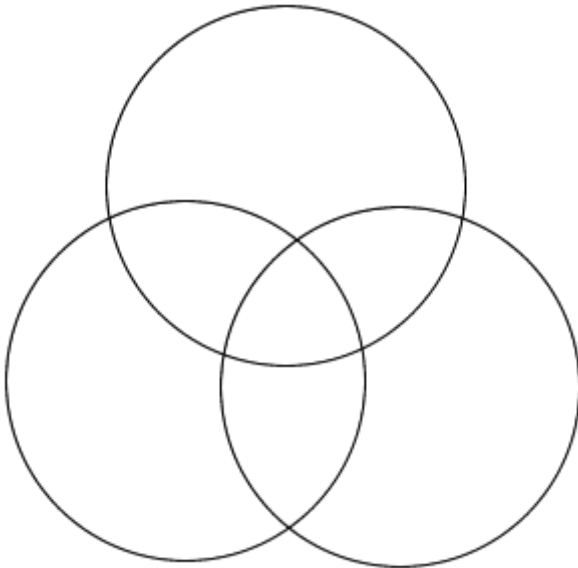
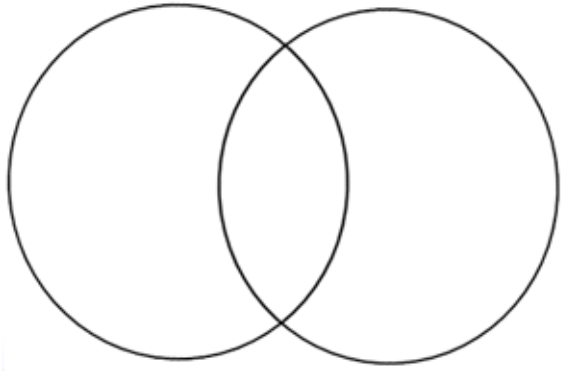
- Data title/label/legend
- Data source (average of what?)
- x: anchor, scale
- y: scale, normalization



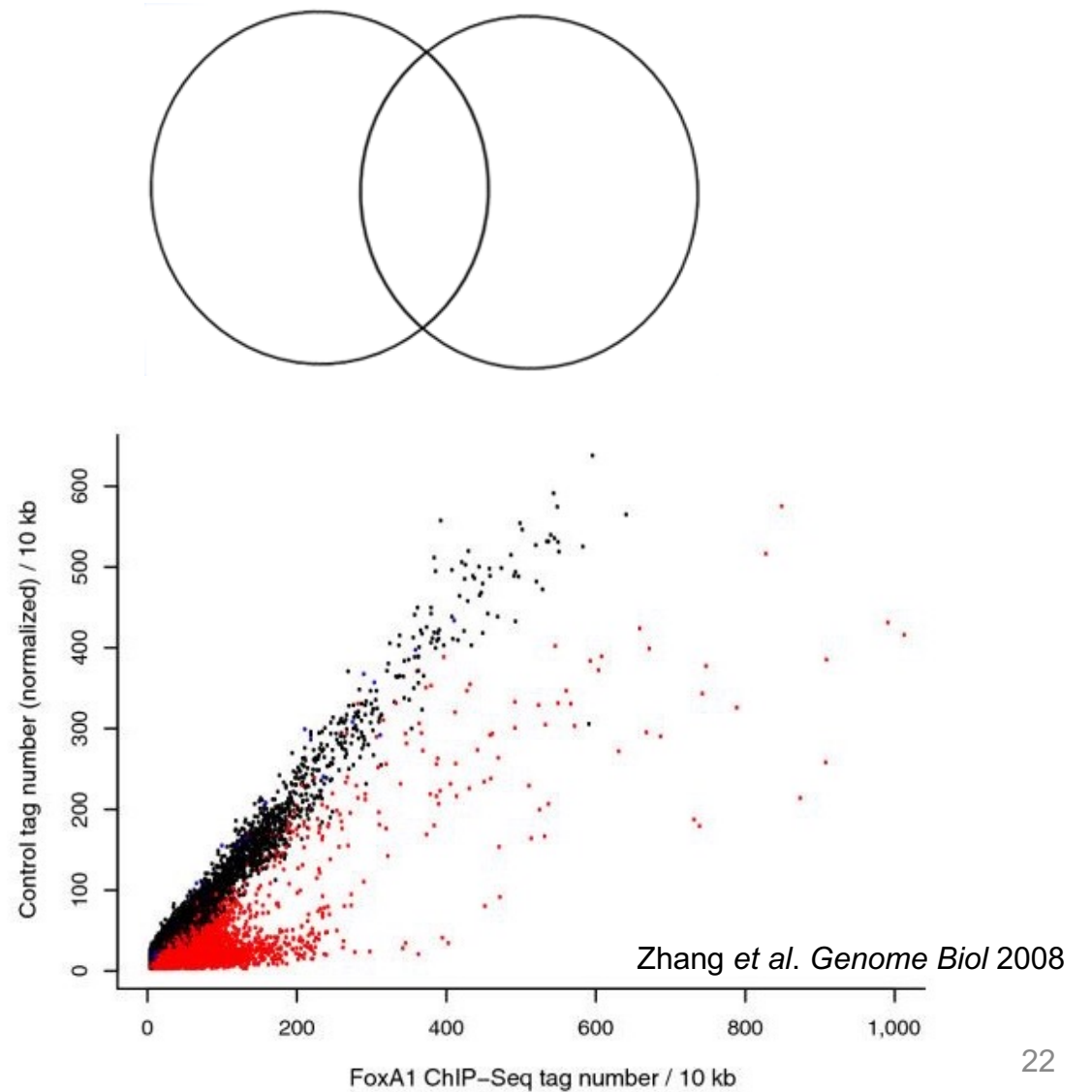
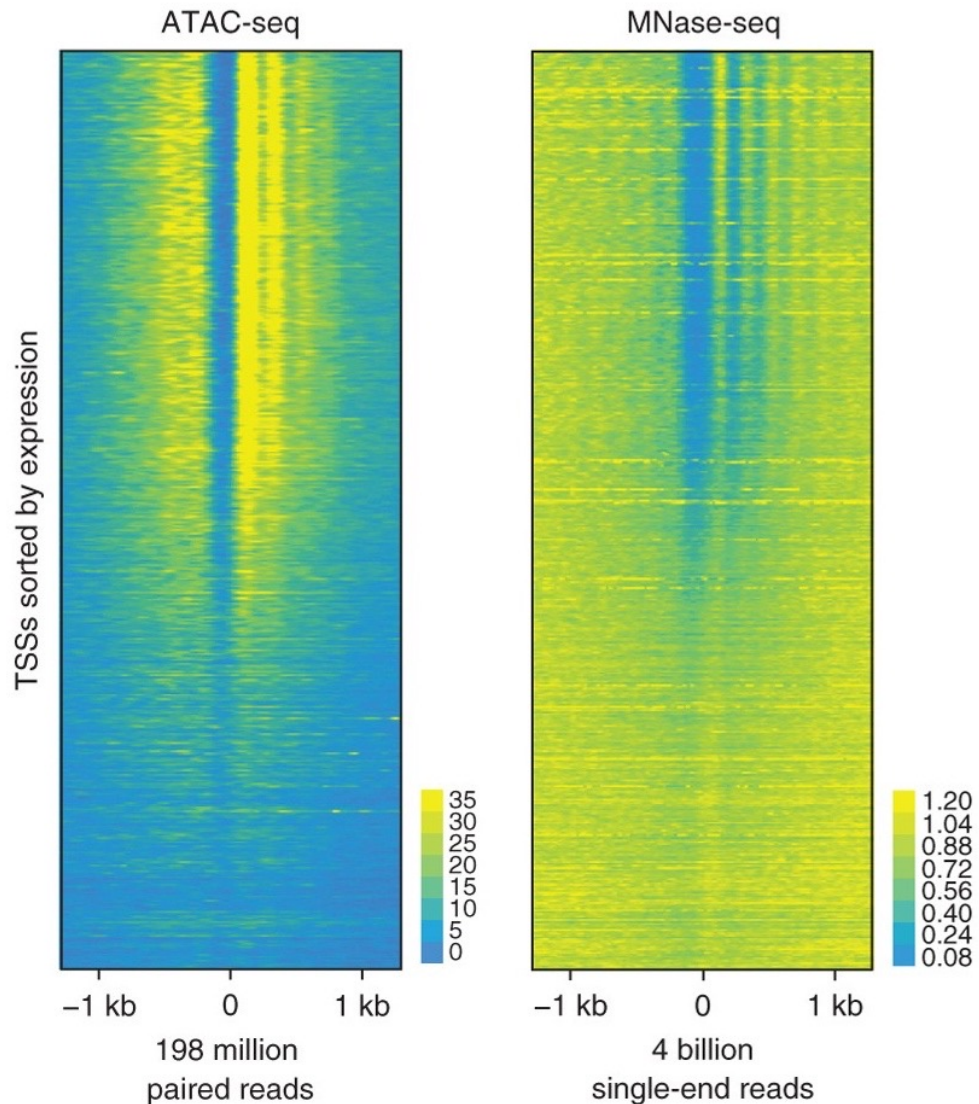
Common misinterpretations of composite plots

- Caveat 1: A peak in a composite plot may be contribute by only a tiny fraction of regions (not representative of global picture)
- Caveat 2: A higher peak does not necessarily mean stronger signal or more region coverage

Venn diagram presents yes/no relations between sets

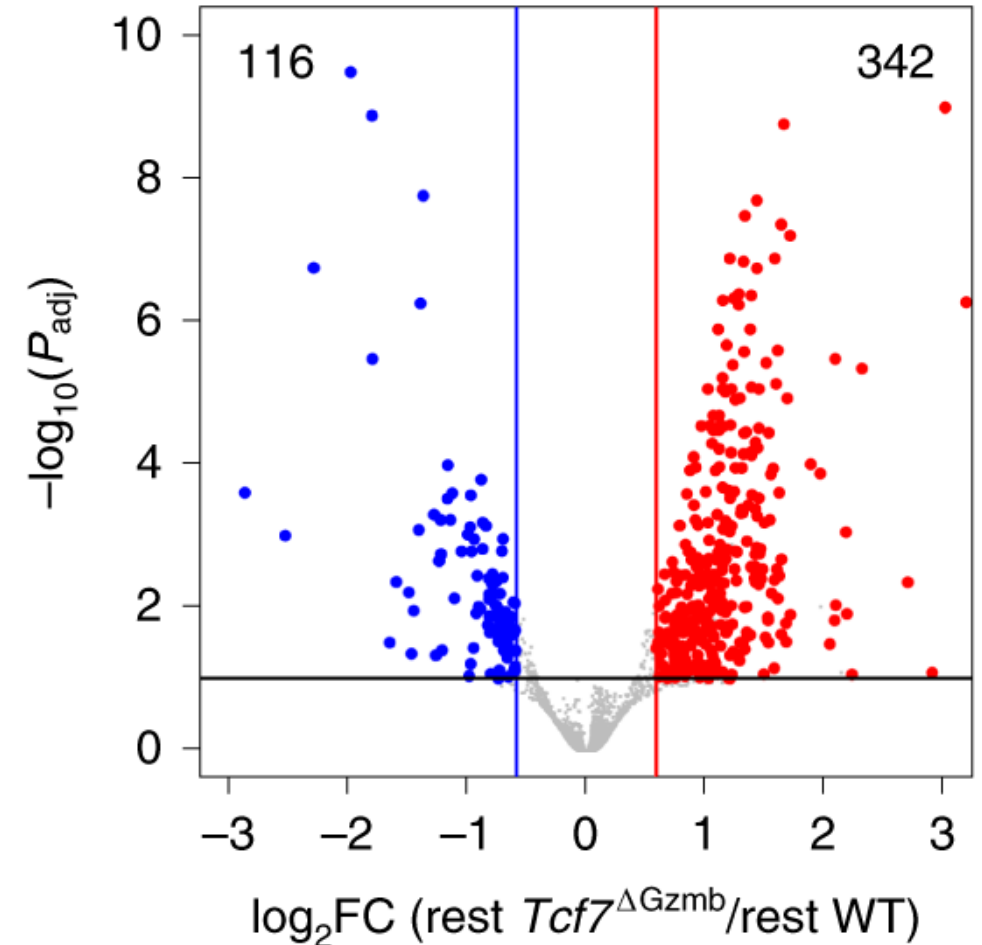


Heatmaps and scatter plots are more informative than a Venn diagram



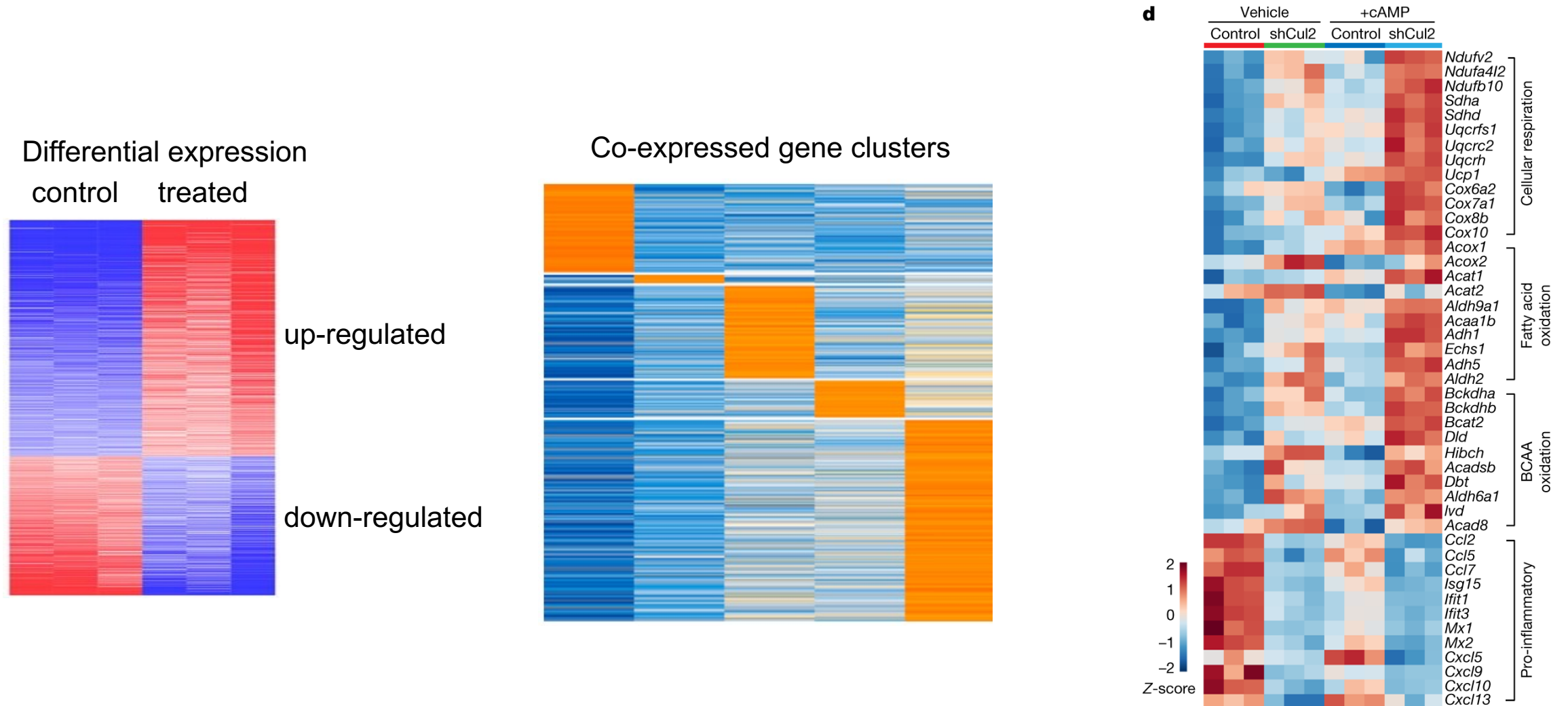
Volcano plot for differential gene expression

- Scatter plot
- 2 dimensions:
 - x: signal strength (e.g., log2 fold change)
 - y: statistical significance (e.g., $-\log_{10}P$)
- Set cutoffs on 2 axes



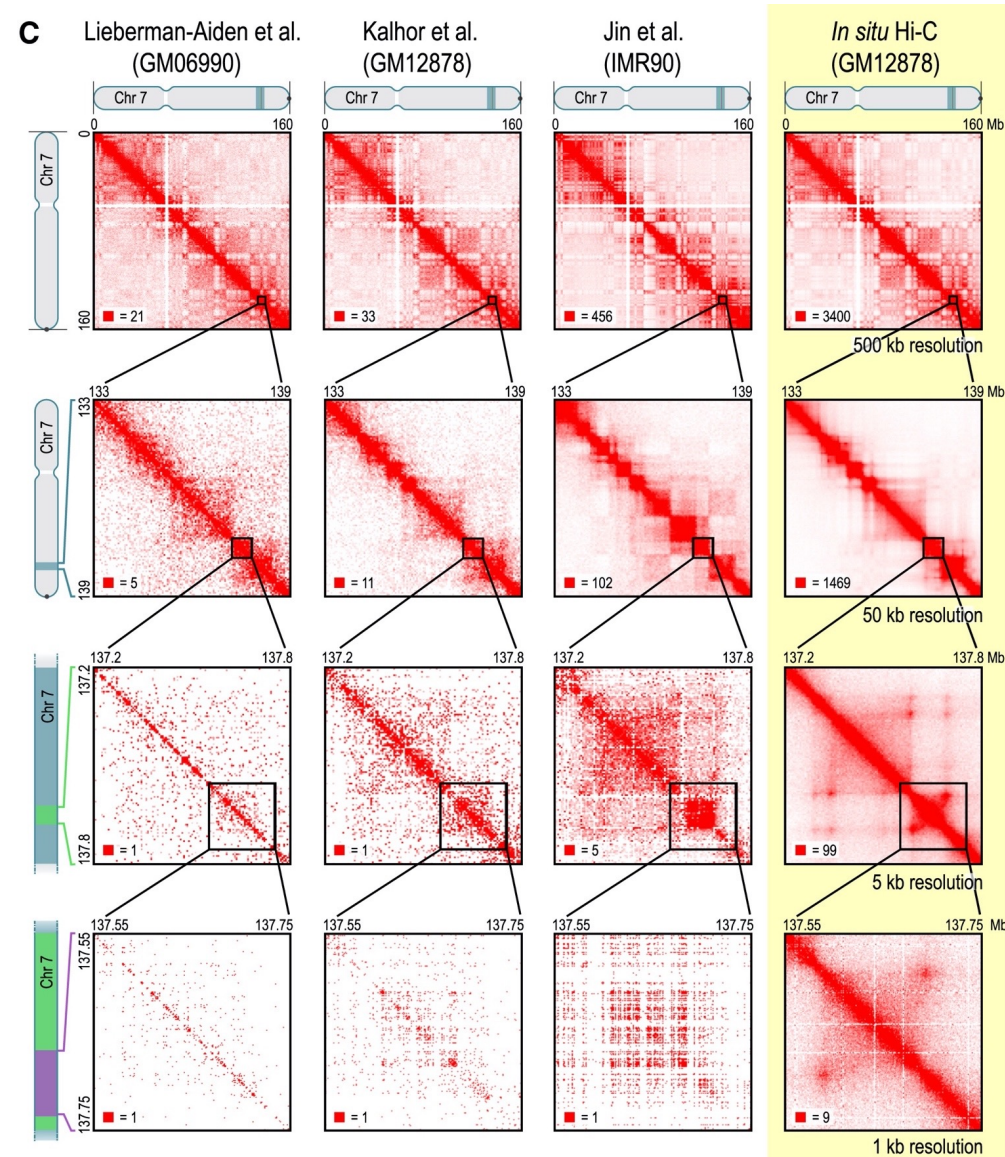
Shan *et al.* *Nature Immunology* 2022

Differential gene expression visualization by heatmap



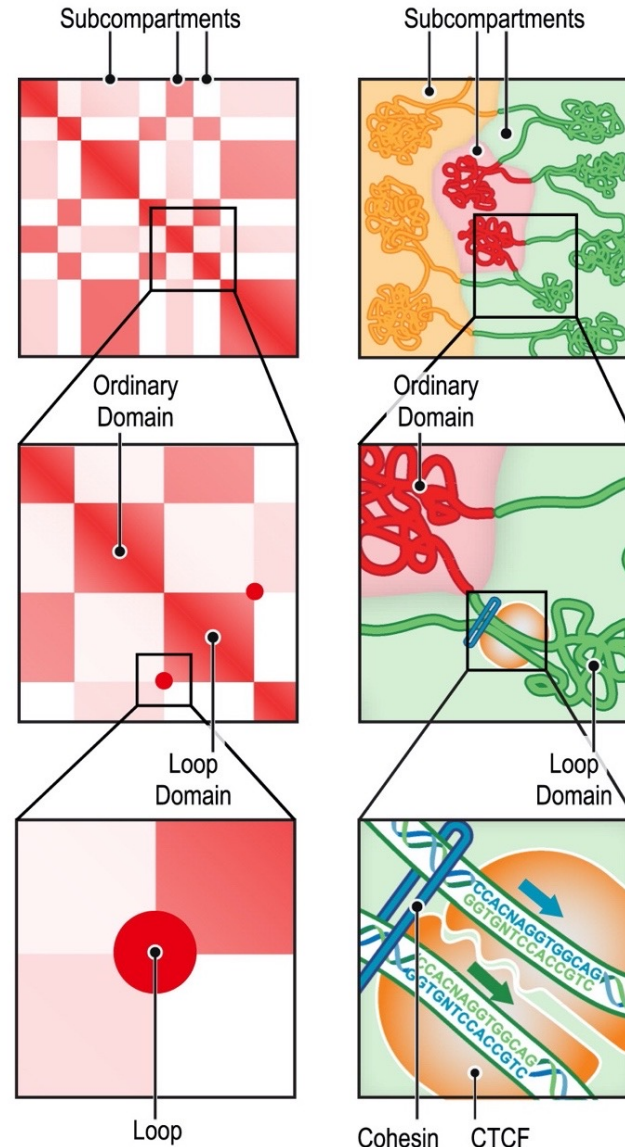
Wang et al. Nature 2022

Hi-C contact heatmap for 3D genome interactions



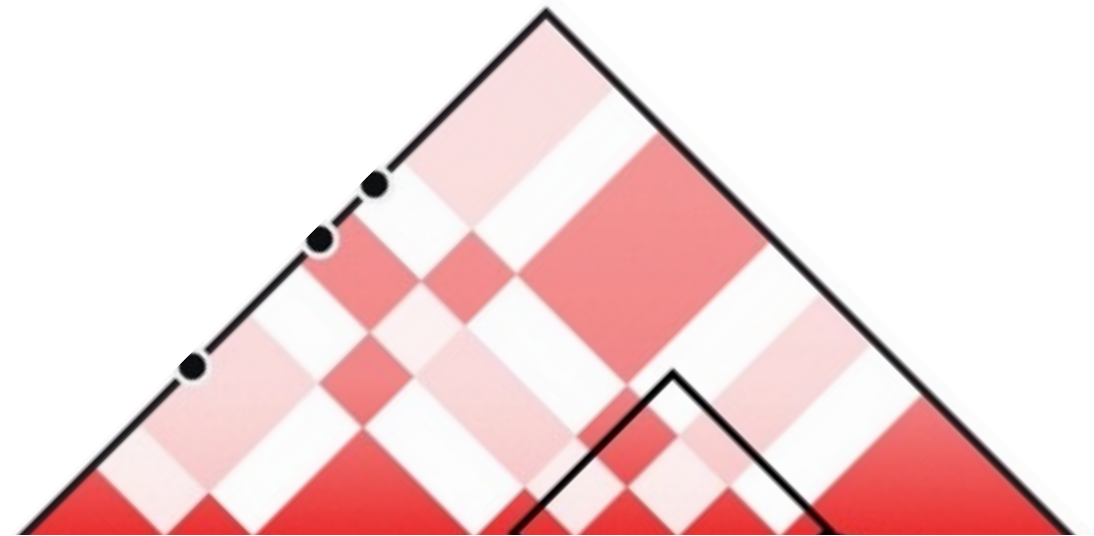
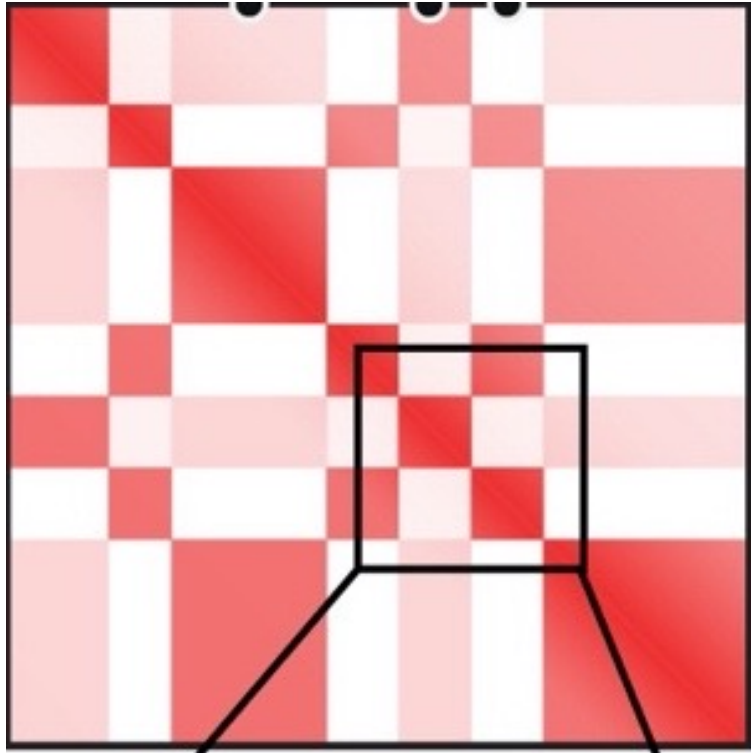
Rao *et al. Cell* 2014

Hi-C contact heatmap for 3D genome interactions



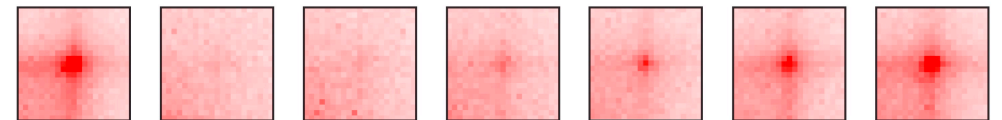
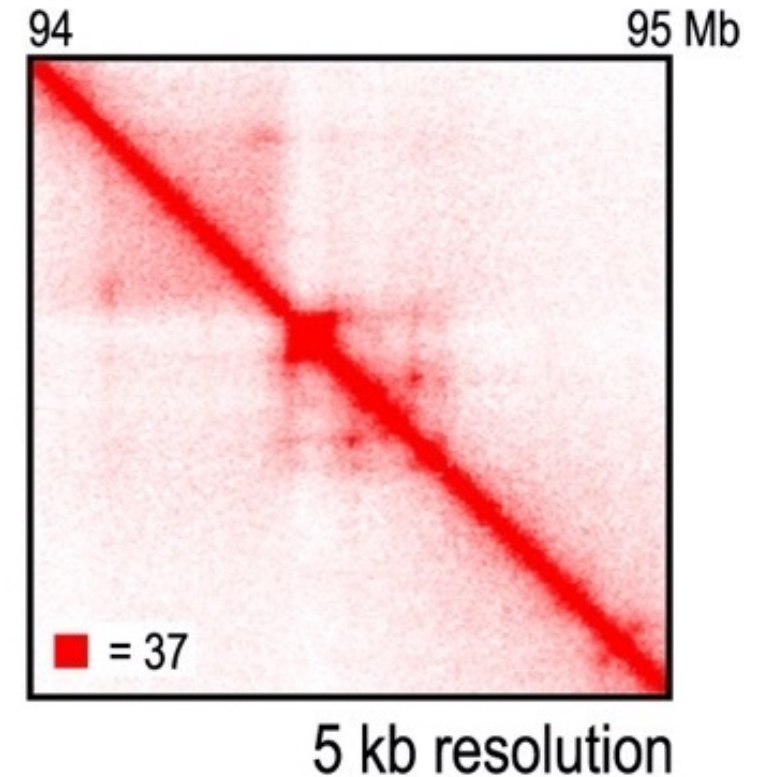
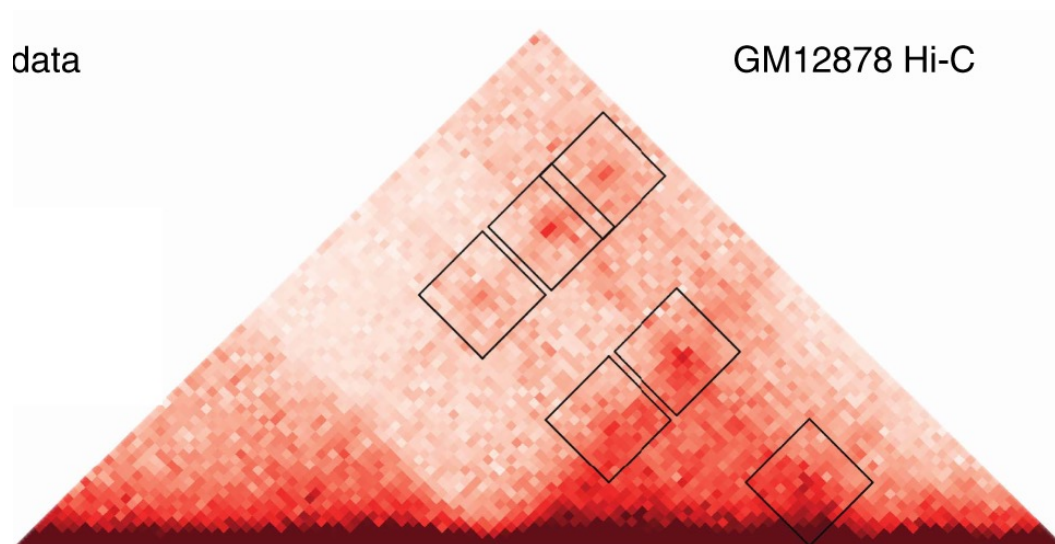
Rao *et al. Cell* 2014

Hi-C contact heatmap for 3D genome interactions

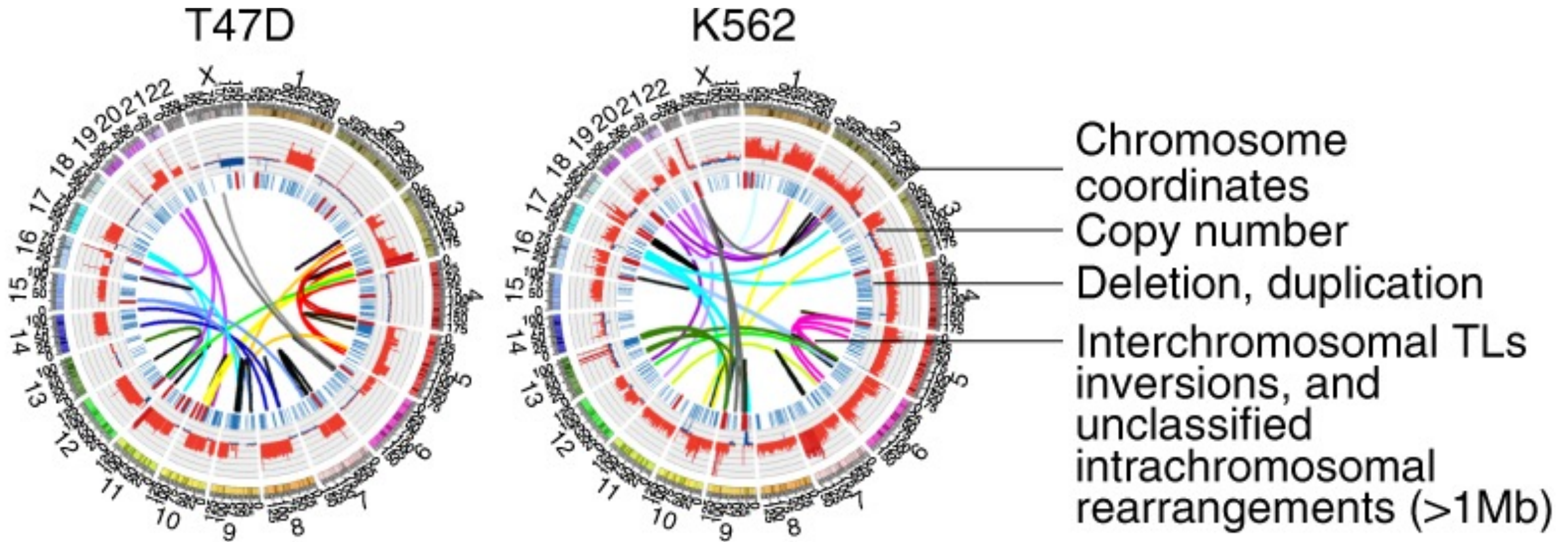


Essential elements in a Hi-C contact heatmap

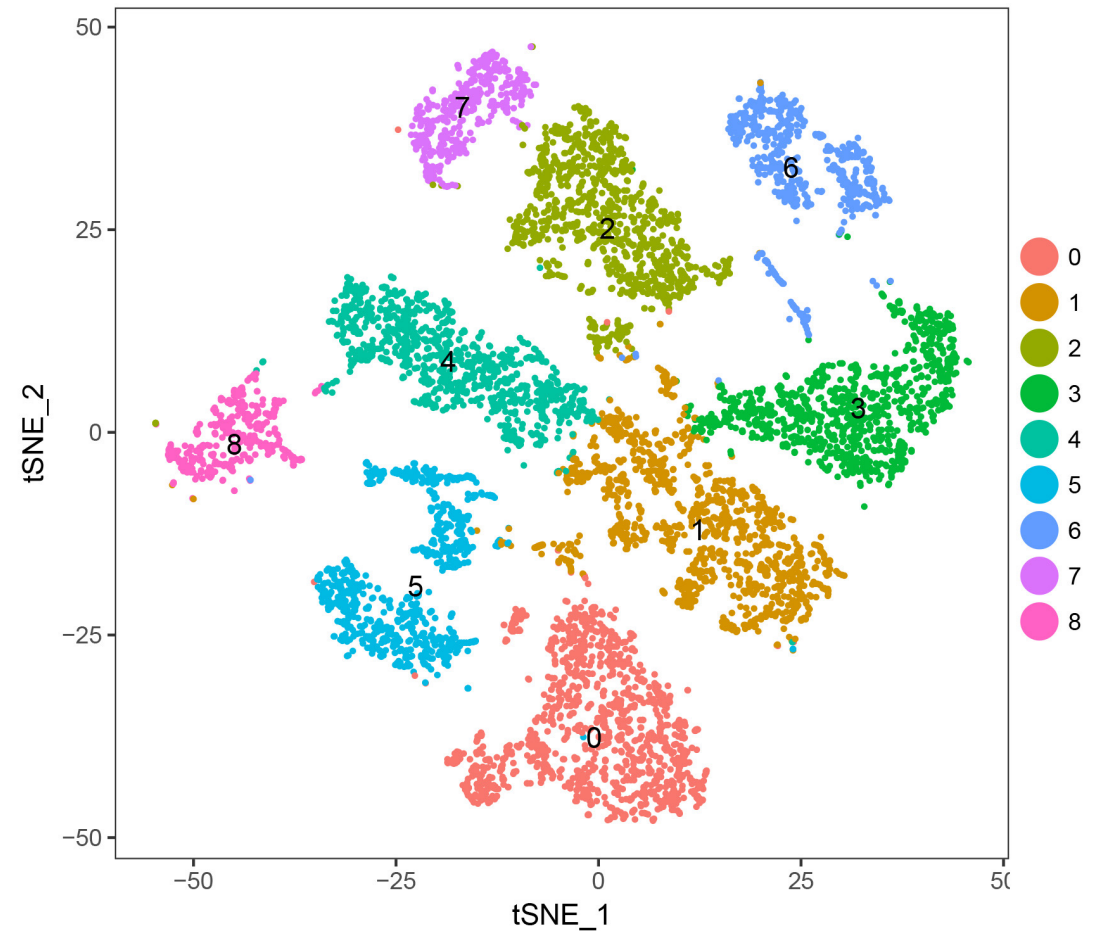
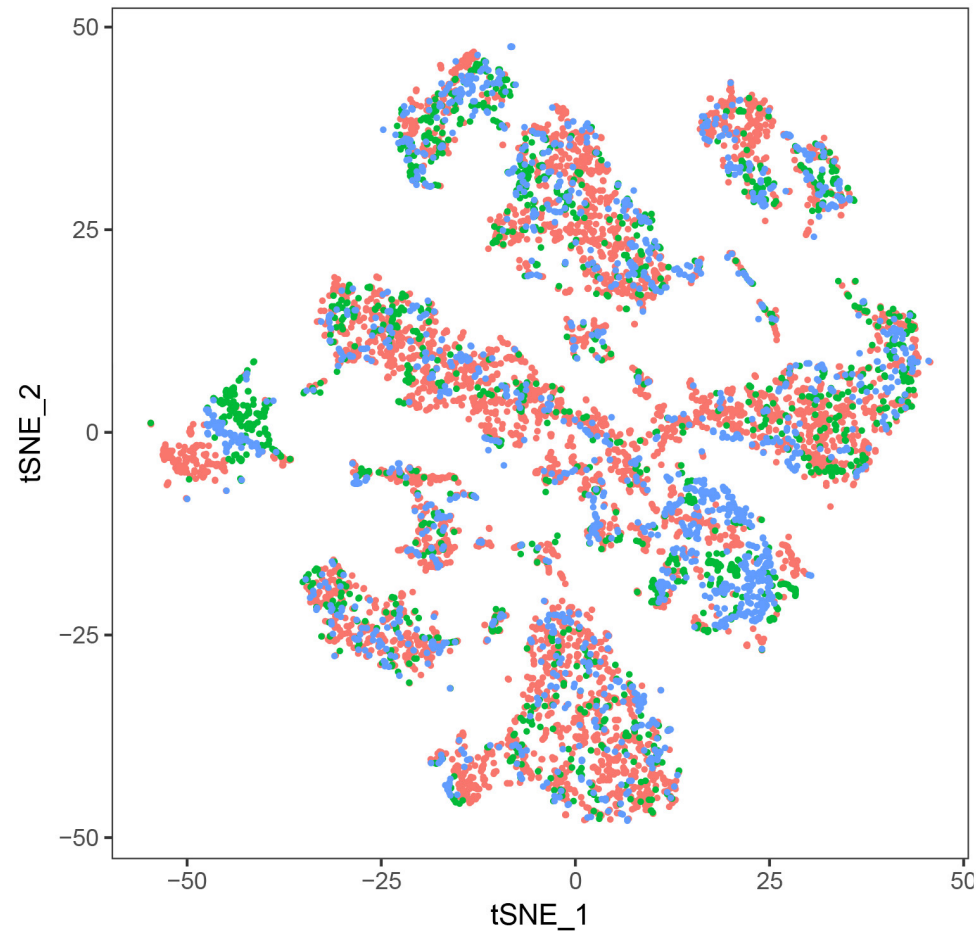
- Scale, scale, scale
- Resolution
- Normalization
- Blocks, stripes, loops (2d peaks)



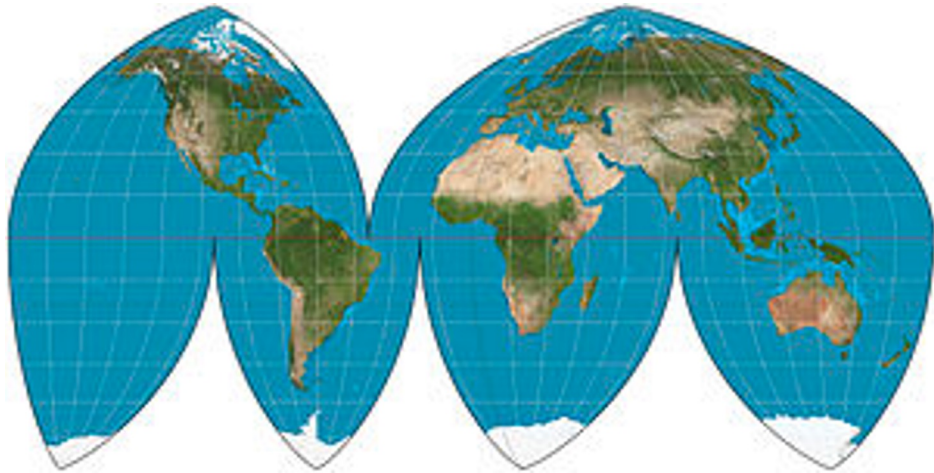
Circos plot integrates multiple layers of genomic information



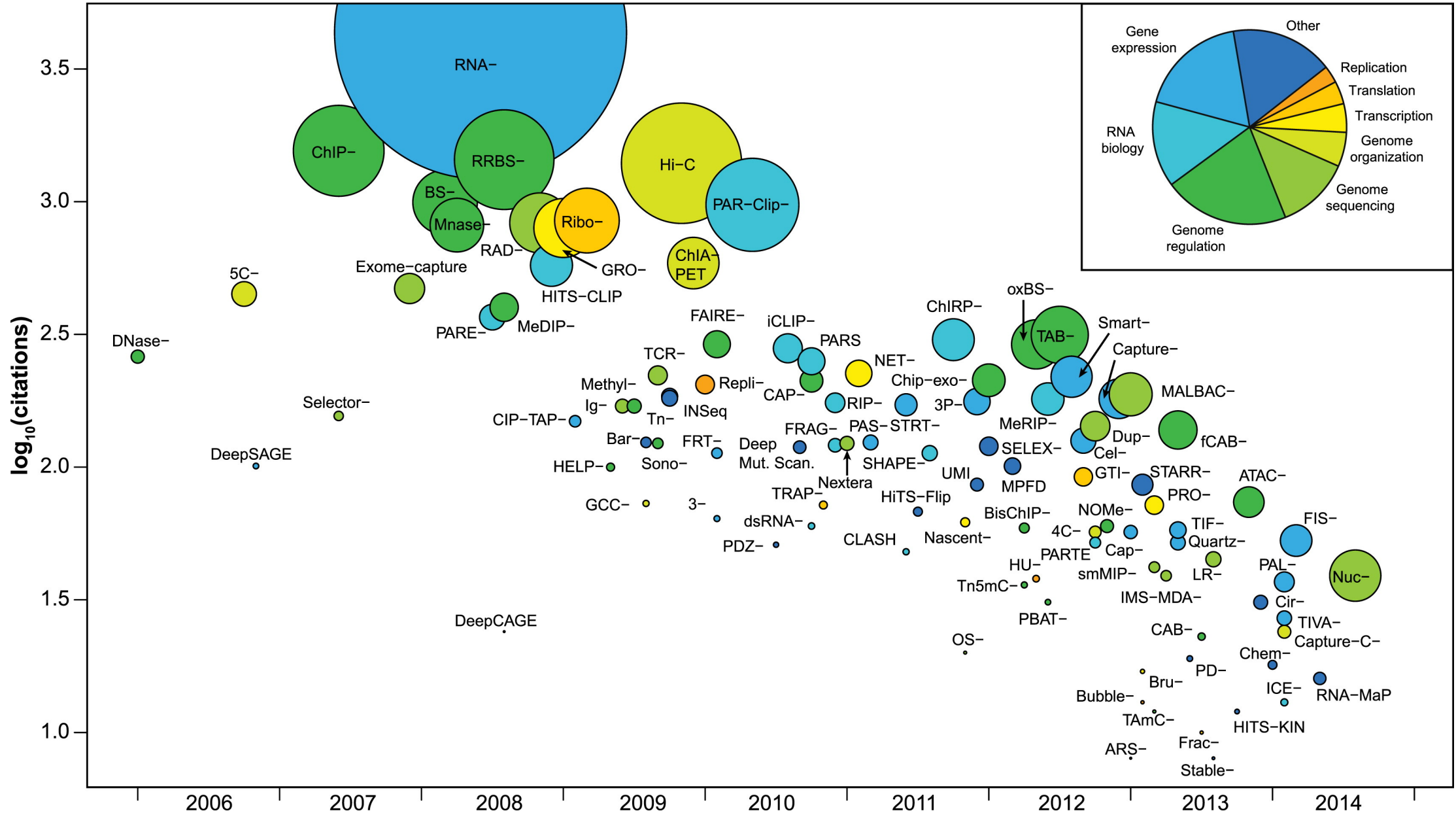
Single-cell data: clustering vs. t-SNE/UMAP visualization



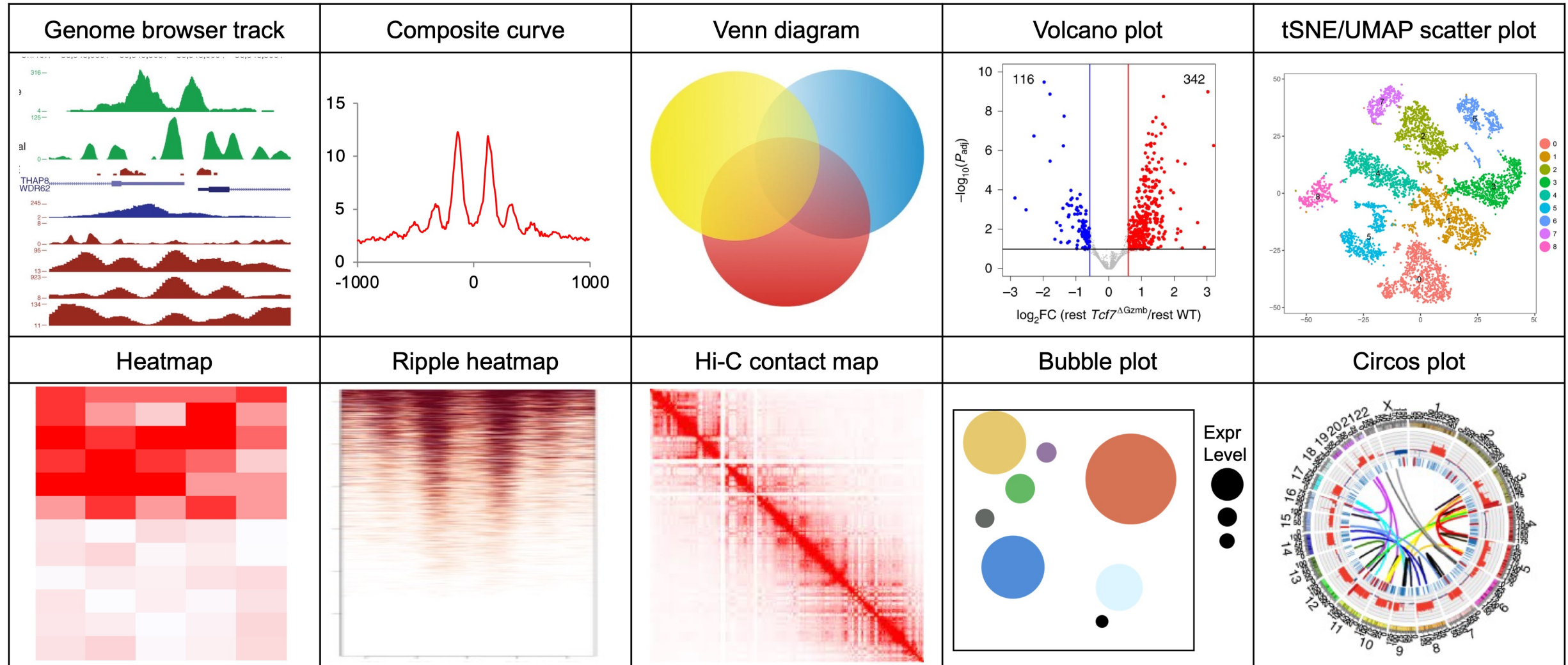
Information may be lost or distorted in dimensionality reduction



Bubble plots: NGS-based applications (-seq)



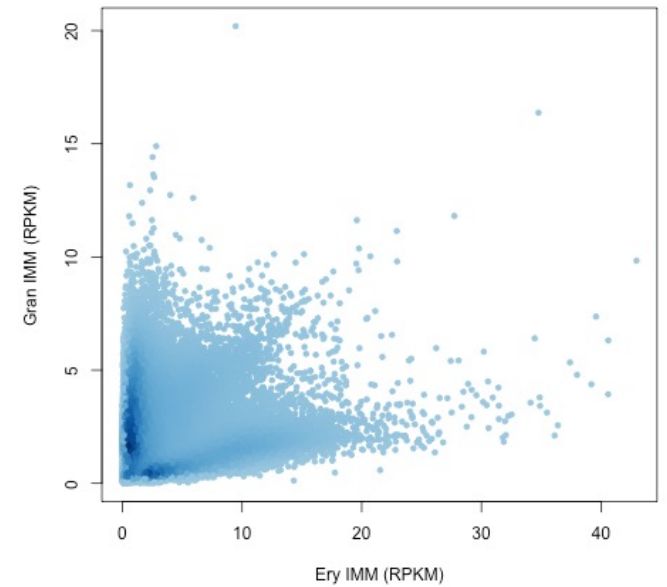
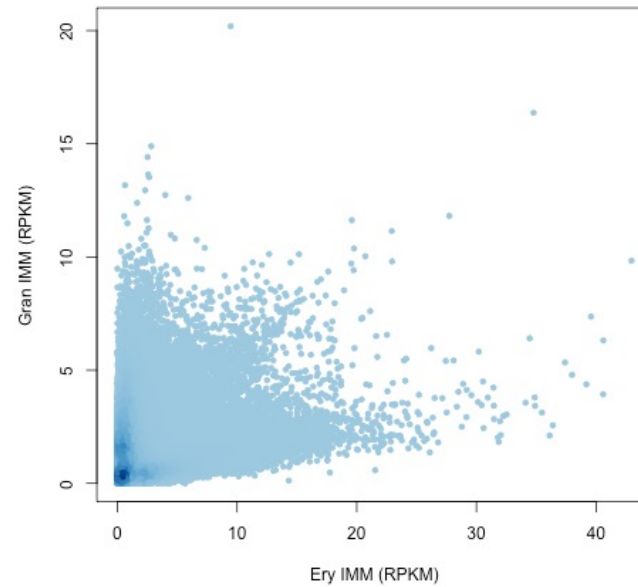
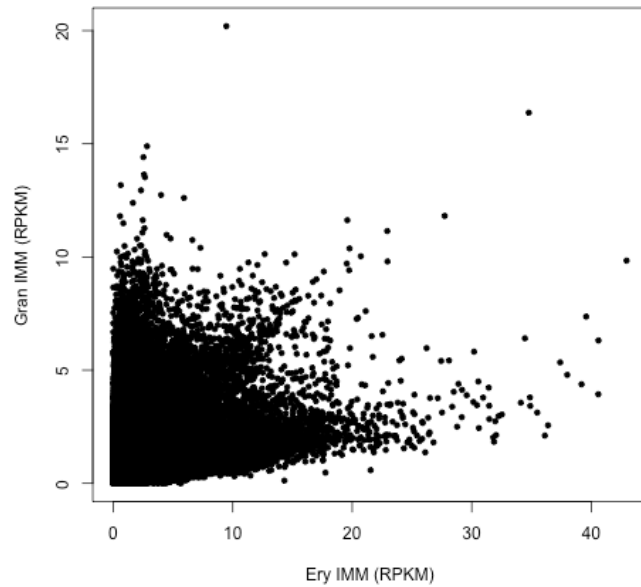
Summary



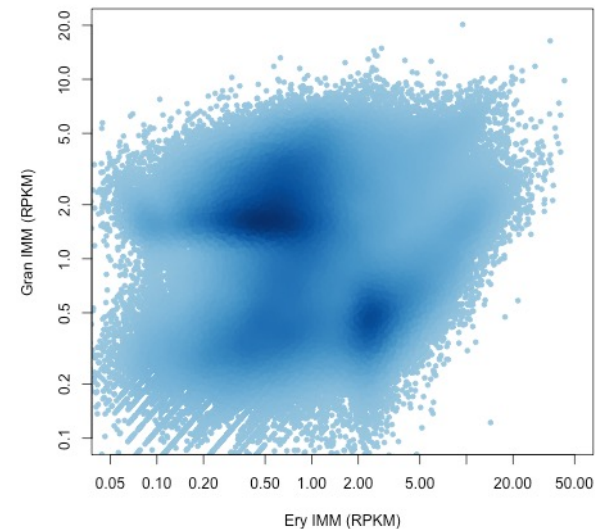
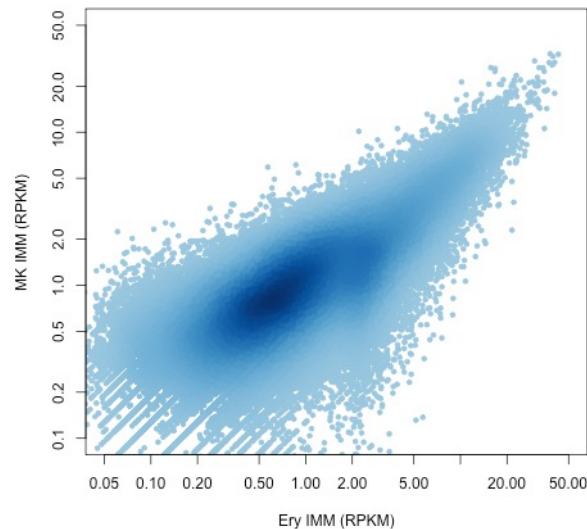
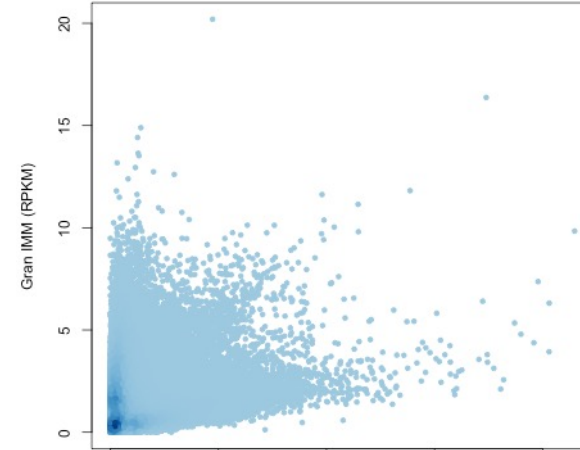
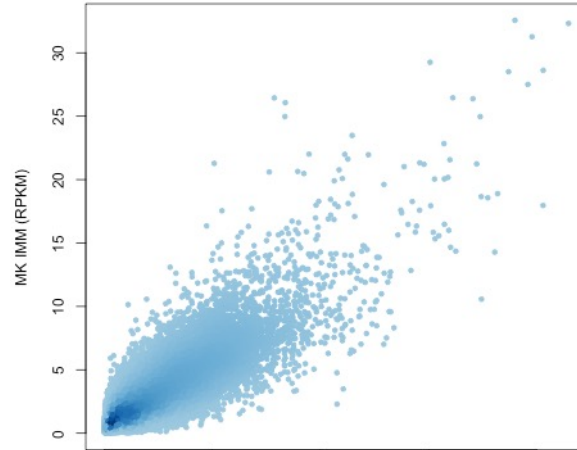
Some tips

Scatter plot

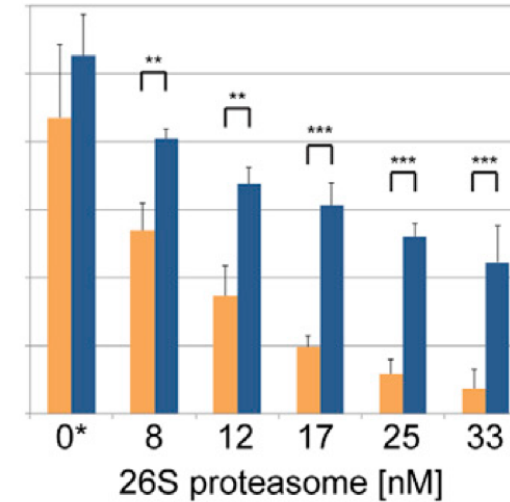
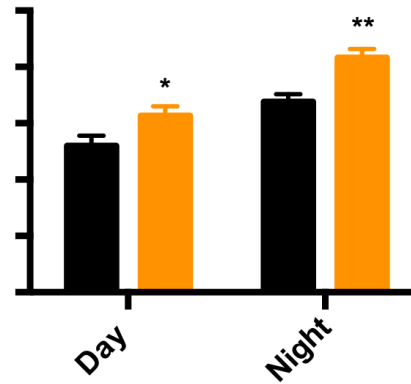
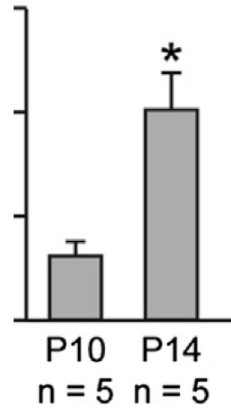
- The density of the data points matters!



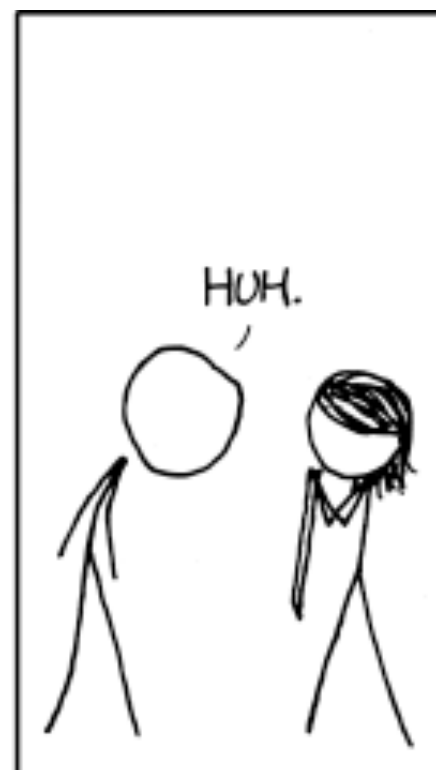
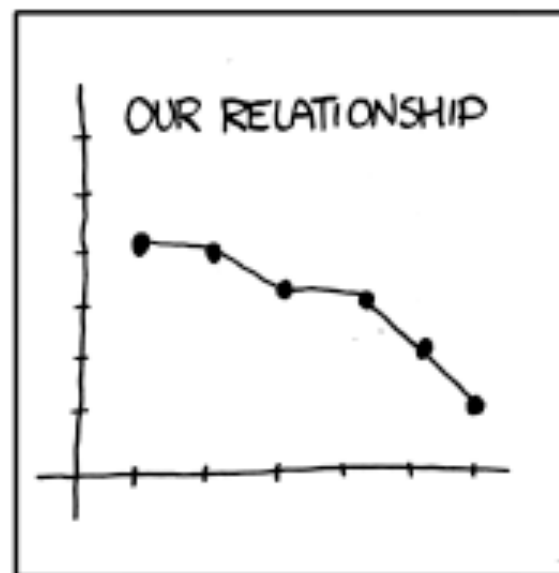
Choose the appropriate axis scale for plotting (linear or logarithm)



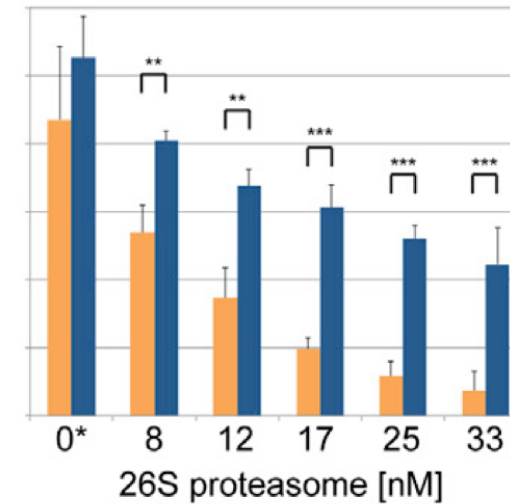
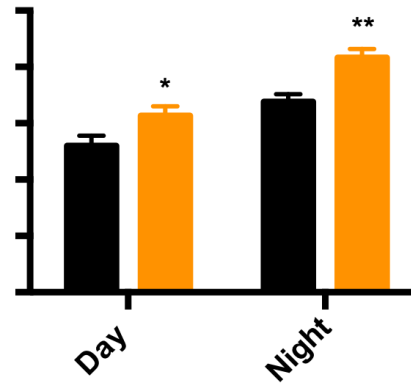
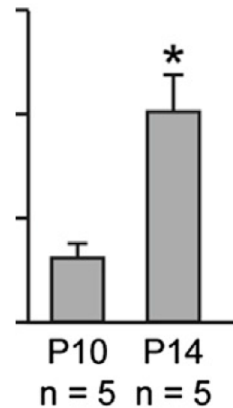
Can you spot any issue with these bar plots?



Plots from *Cell* 2014



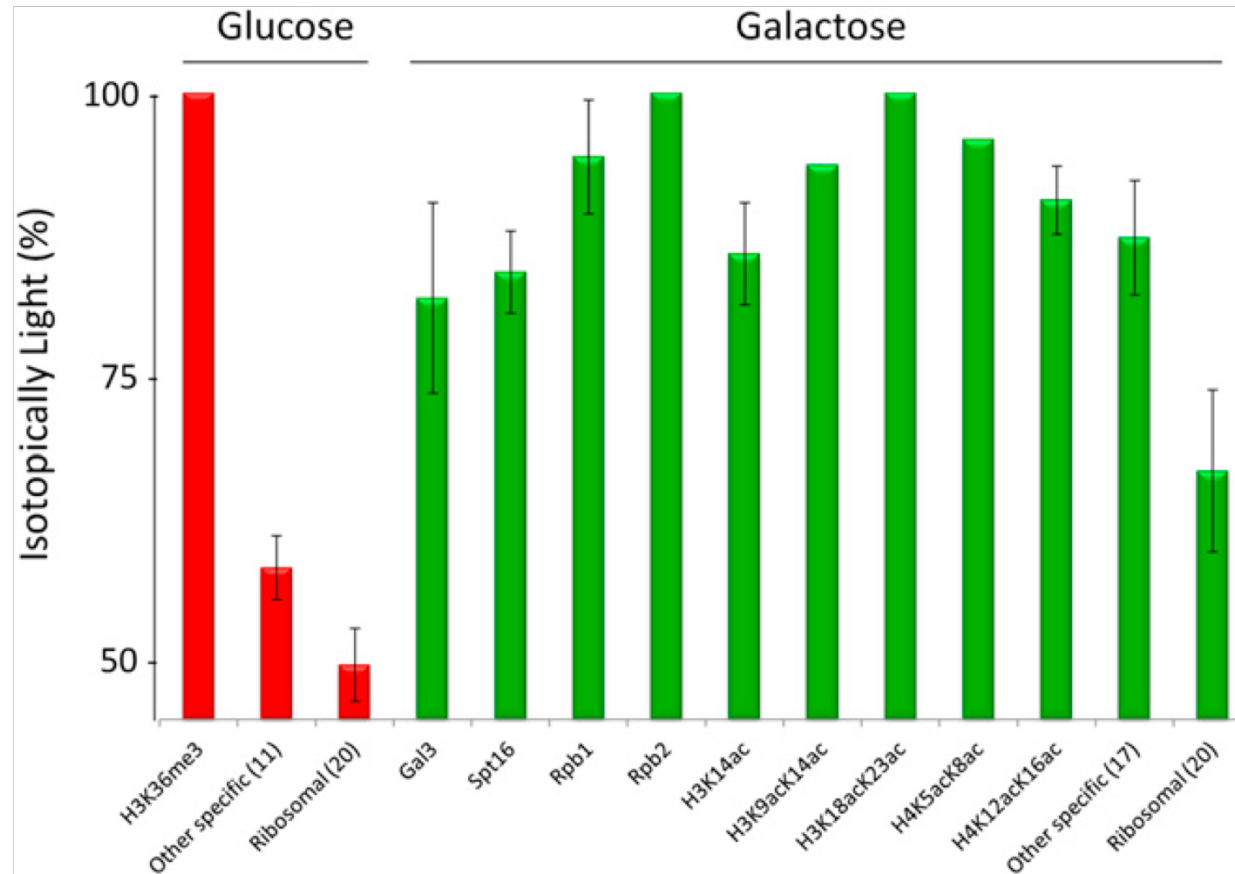
Can you spot any issue with these bar plots?



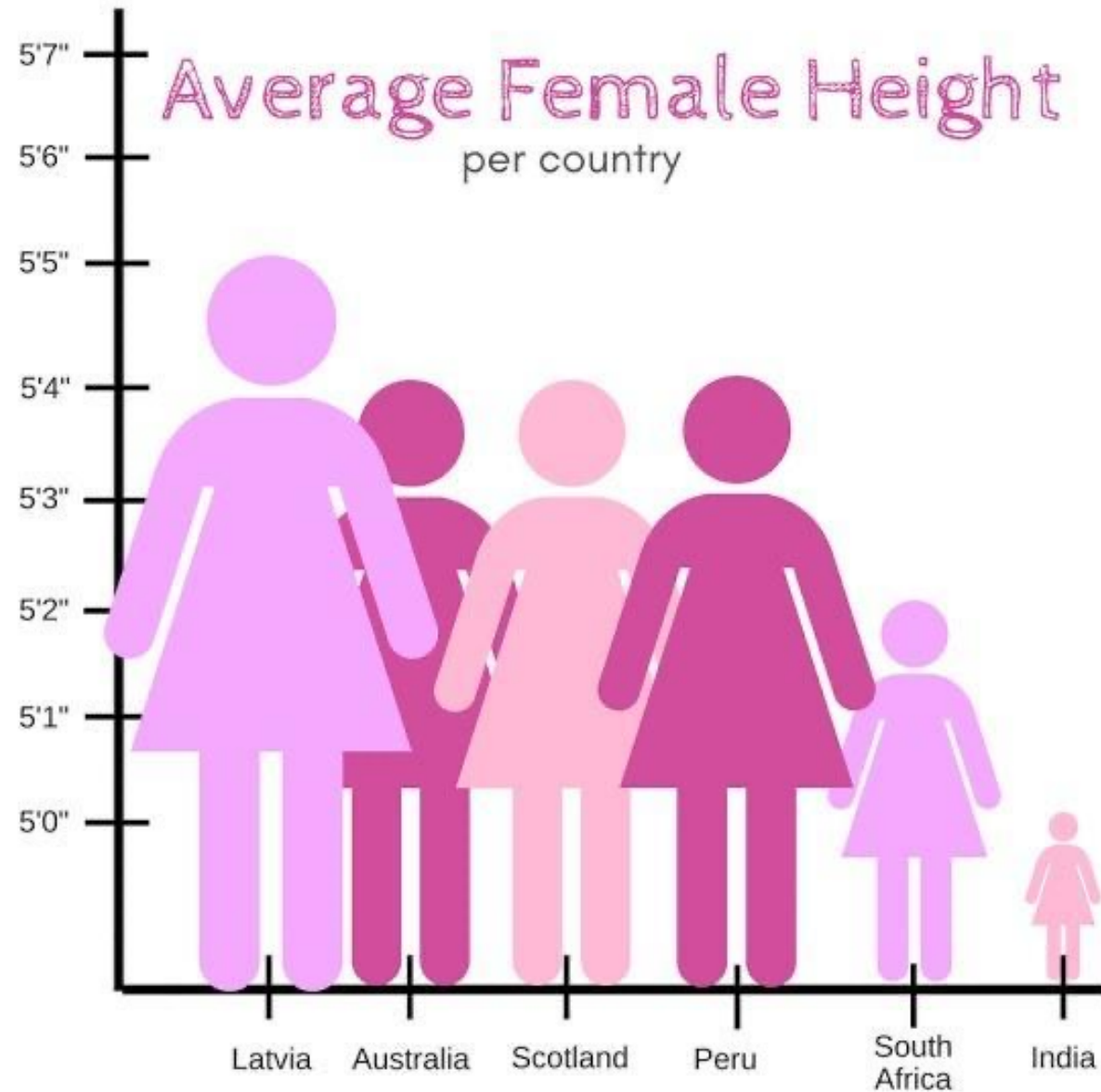
Bar chart y-axis should always start at 0, and on a linear scale.
If the difference is small, box plots or original data points are better.

Plots from *Cell* 2014

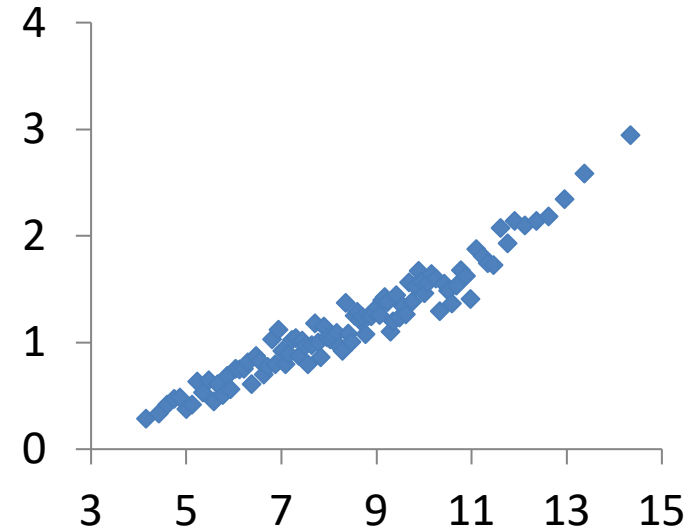
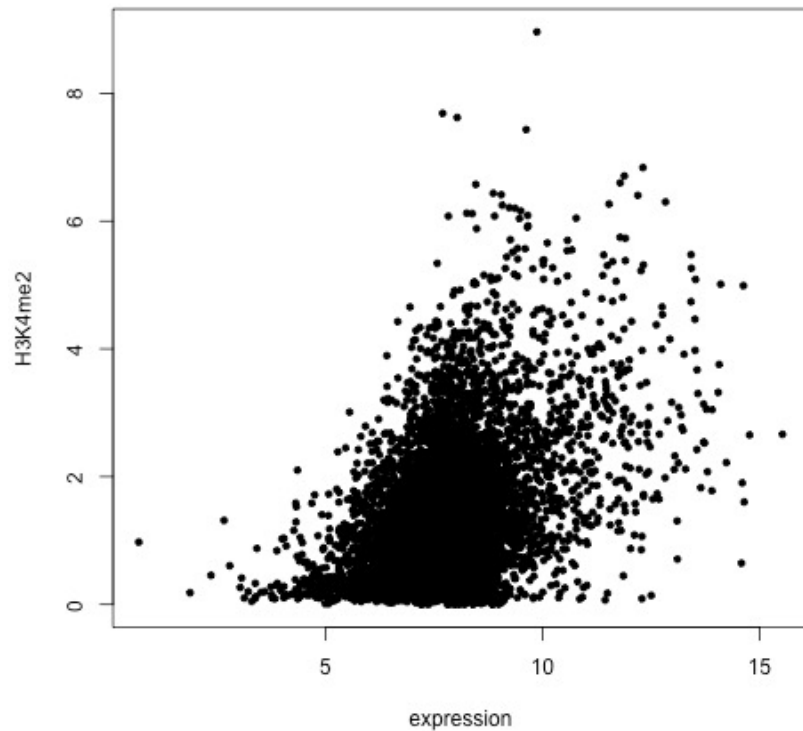
Another example



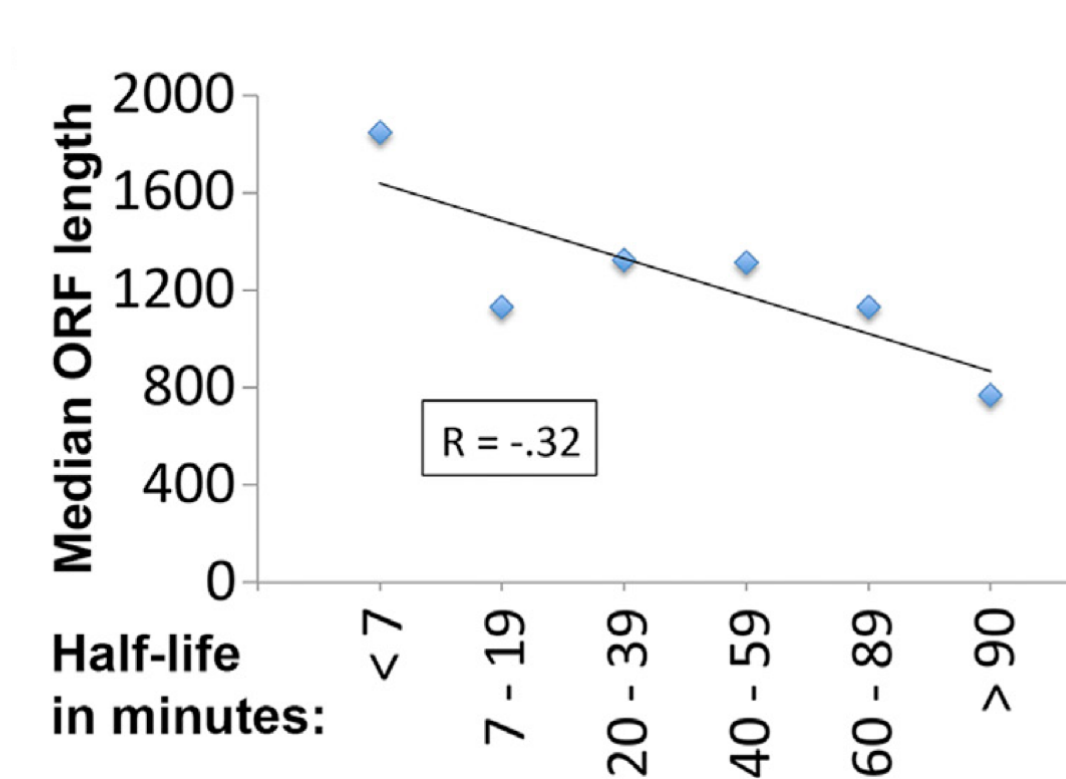
Byrum et al. *Cell Reports* 2012



Scatter plot? Group them if needed

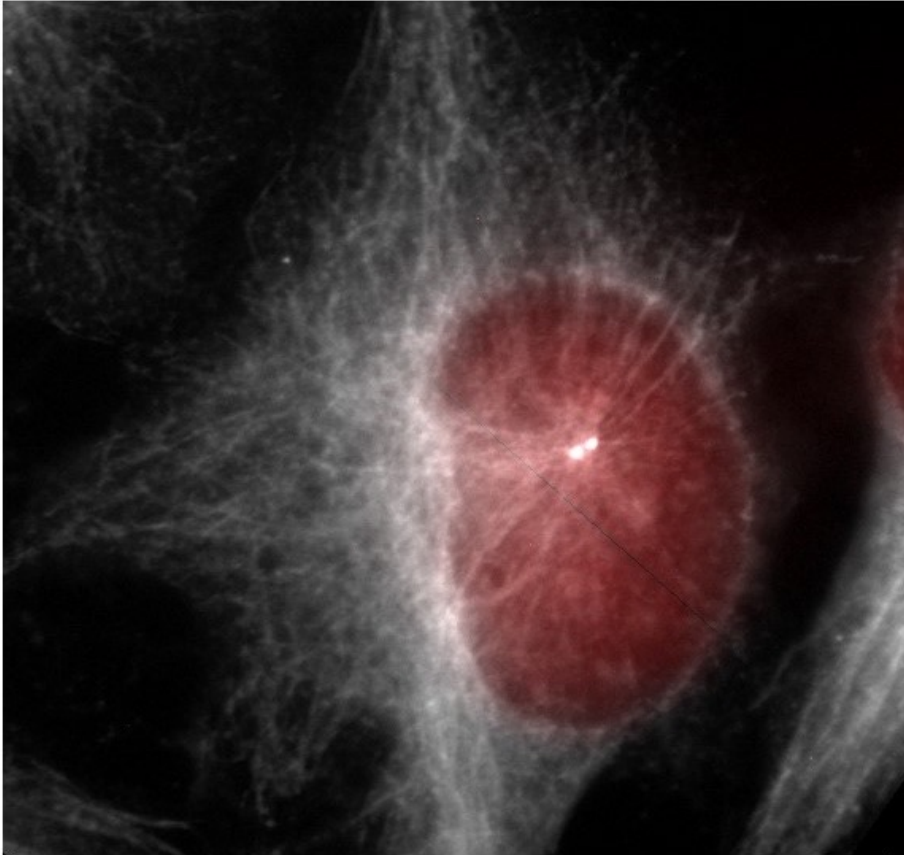


However, grouping should be fair/unbiased

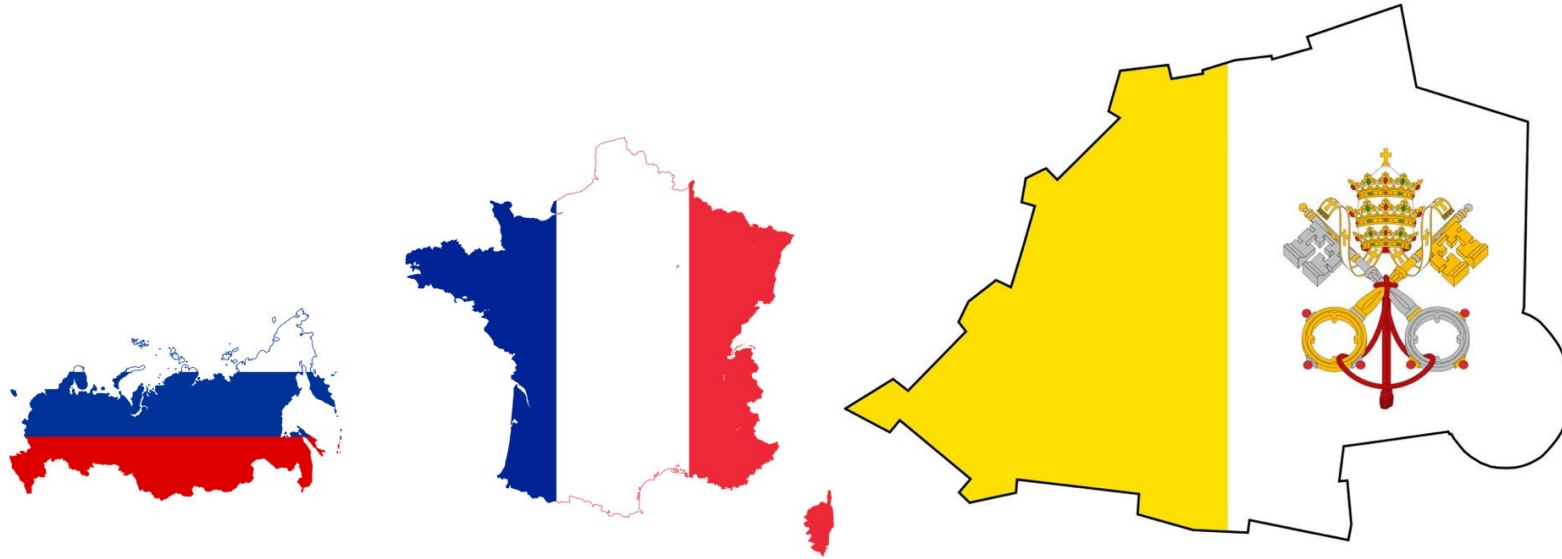


Cell 2014

Scale matters!



Areas of Russia, France and Vatican compared*



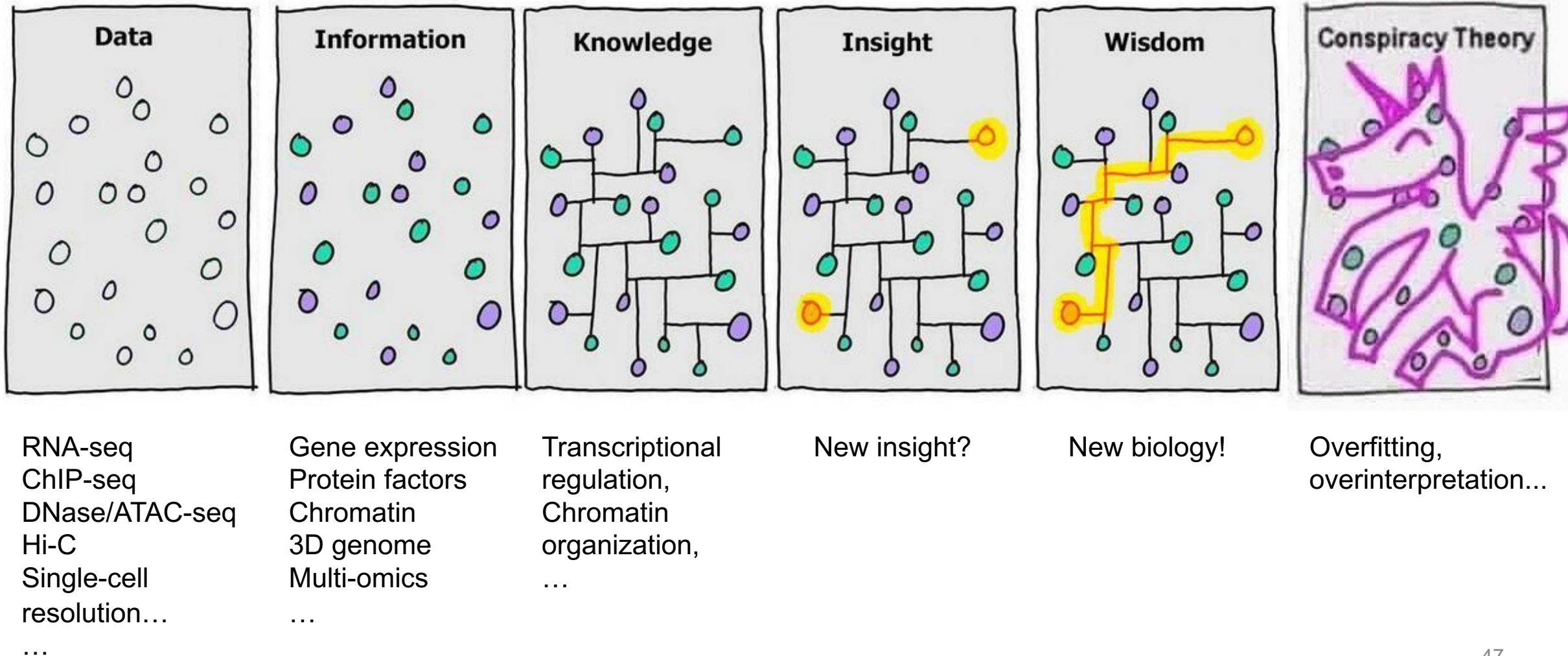
*not to scale

Scale matters. It does!

Take-Home Tips

- Always check the axes and pay attention to the scale.
- Bar charts should always start at zero, on a linear scale.
- Pay attention to data point density in scatter plots.
- Group data points when needed, but do it in an unbiased way.

Having the data is not enough; presentation and interpretation matter



HOW TO: DRAW A HORSE

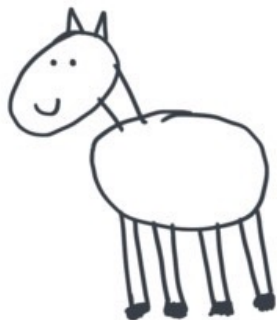
BY VAN OKTOP



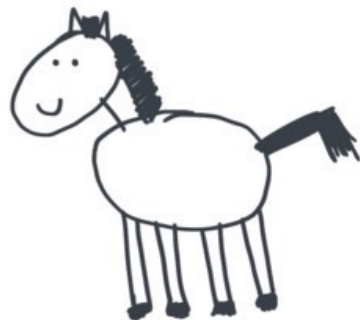
① DRAW 2 CIRCLES



② DRAW THE LEGS

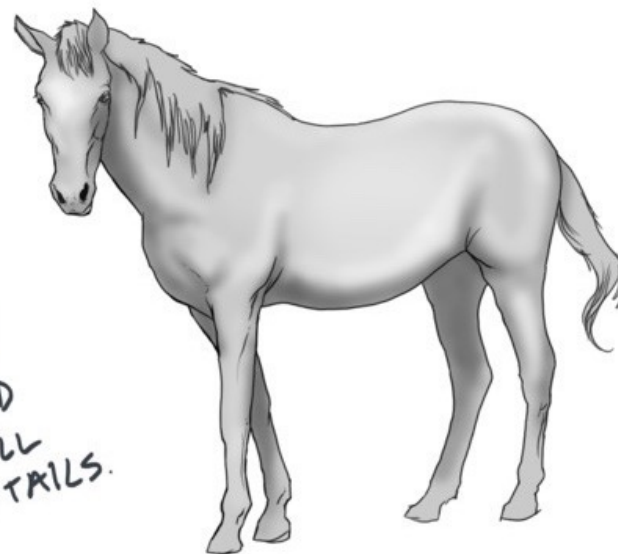


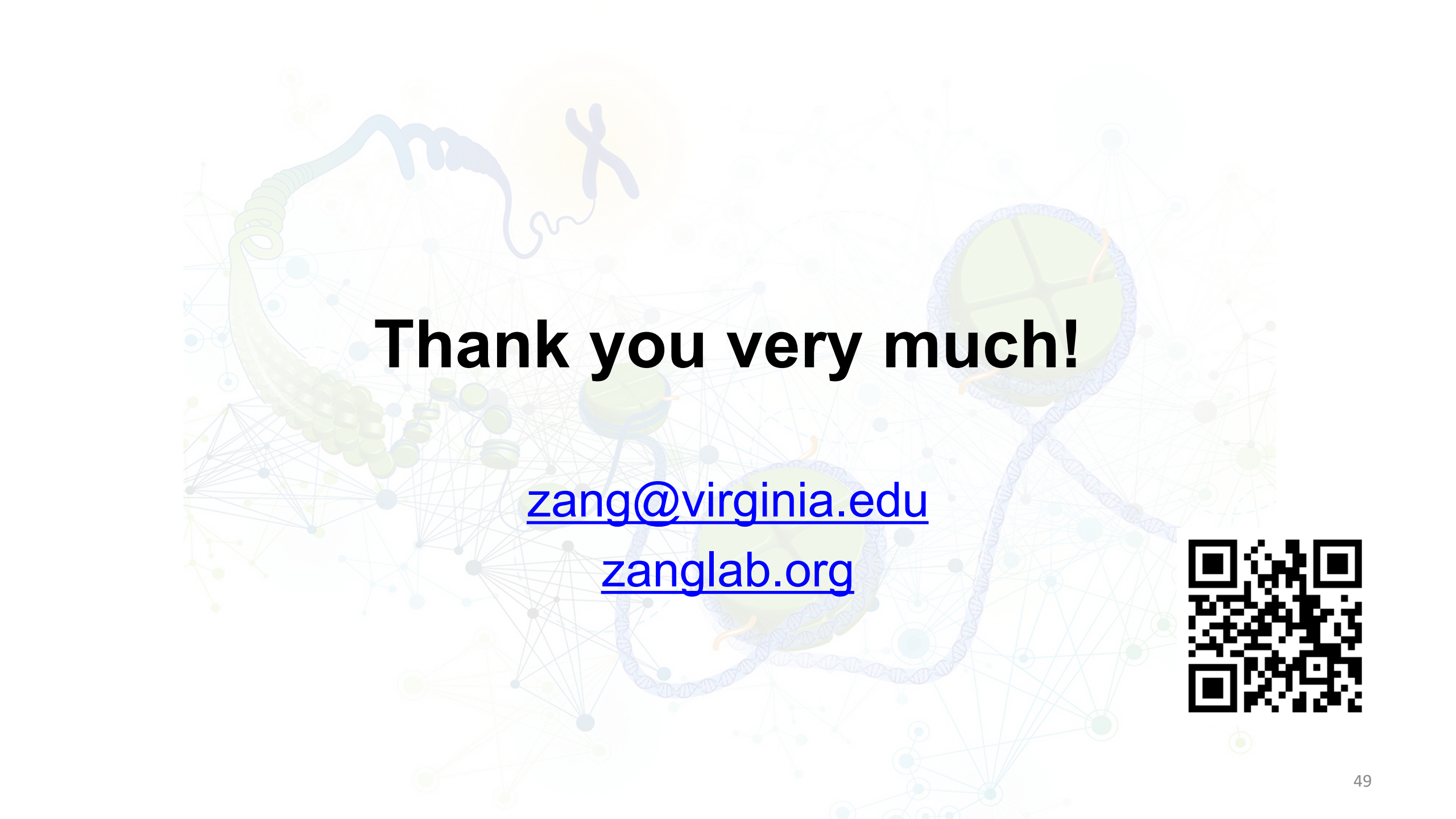
③ DRAW THE FACE



④ DRAW THE HAIR

⑤
ADD
SMALL
DETAILS.





Thank you very much!

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